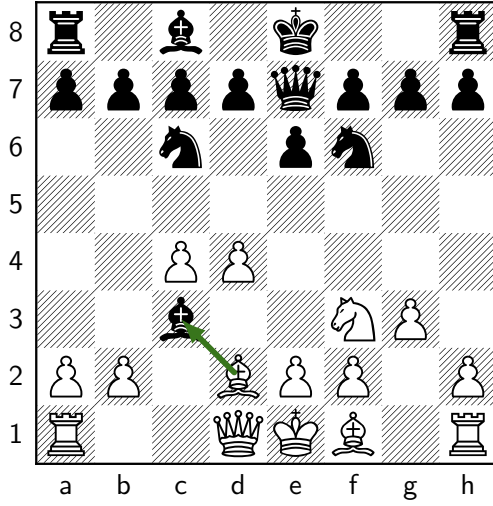


Melanie-MASTERS (human masters) — Bogo-Indian

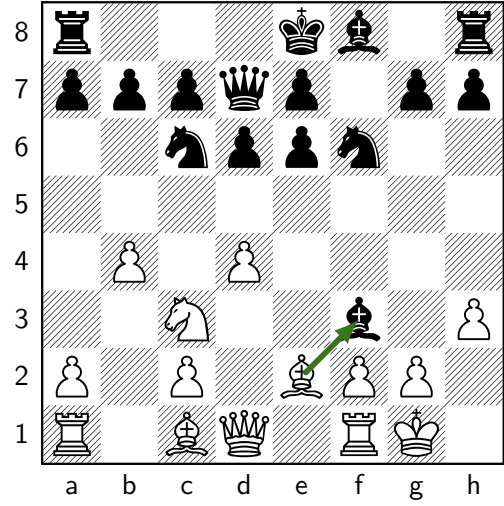
Melanie (Deep_Vision)

Abstract

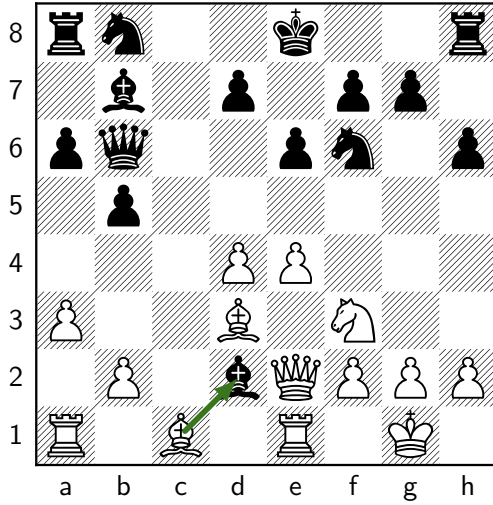
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



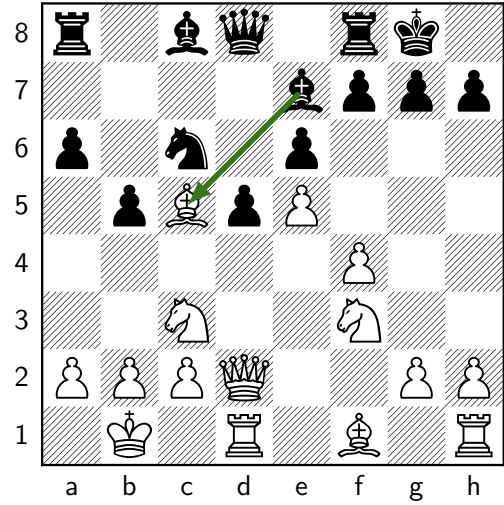
(a)



(b)

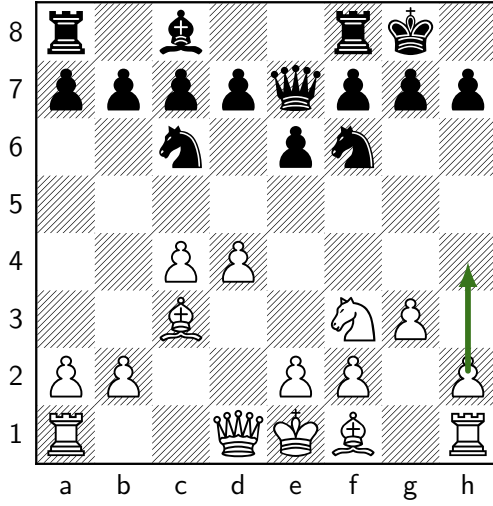


(c)

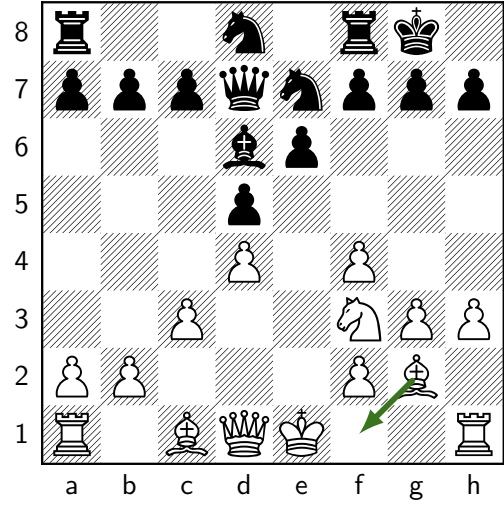


(d)

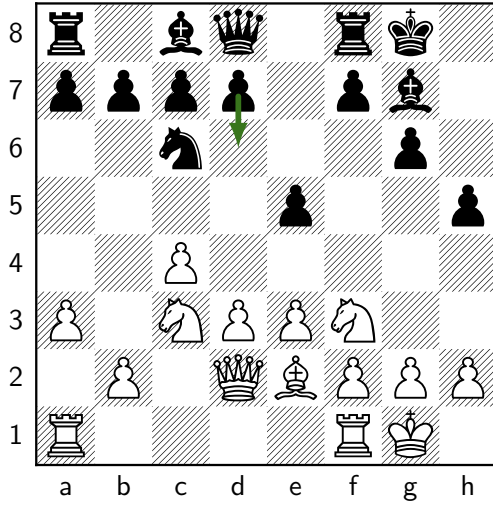
Figure 1: **Move 7, White: Bxc3.** (a) Our position; Melanie plays Bxc3 (d2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#2850` (ply 20) [structural distance $d=71$, QAP obj=128]: the master played Bxf3; under μ our Bd2 plays the role of that game’s Be2; cross-type roles: our Pe2 as their Rf1, our Rh1 as their Pa2; transferred move d2c3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#3074` (ply 22) [structural distance $d=109$, QAP obj=65]: the master played Bxd2; under μ our Bd2 plays the role of that game’s Bc1; cross-type roles: our Nf3 as their Ra1, our Bd2 as their Re1; transferred move d2c3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#2817` (ply 23) [structural distance $d=103$, QAP obj=37]: the master played Bxc5; under μ our Bd2 plays the role of that game’s Be2; cross-type roles: our Pc4 as their Ra8, our Pd4 as their Qd8; transferred move d2c3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bxc3 (summed QAP objective 281, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



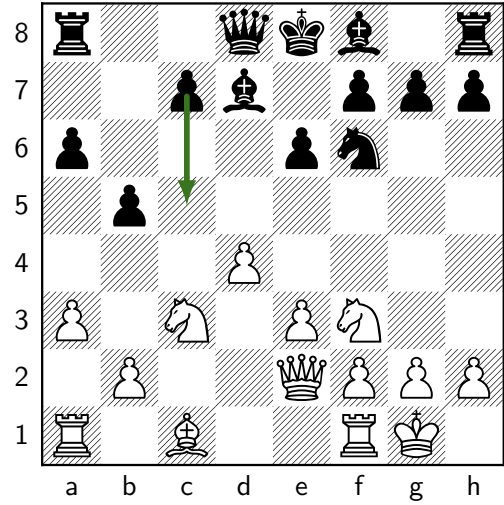
(a)



(b)

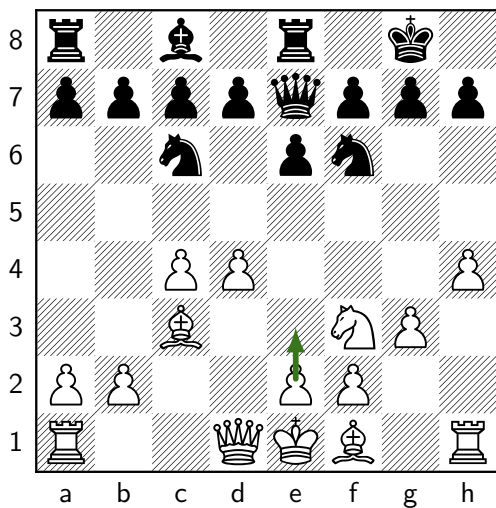


(c)

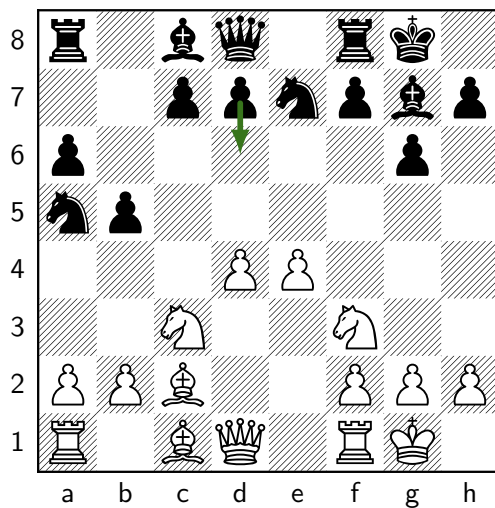


(d)

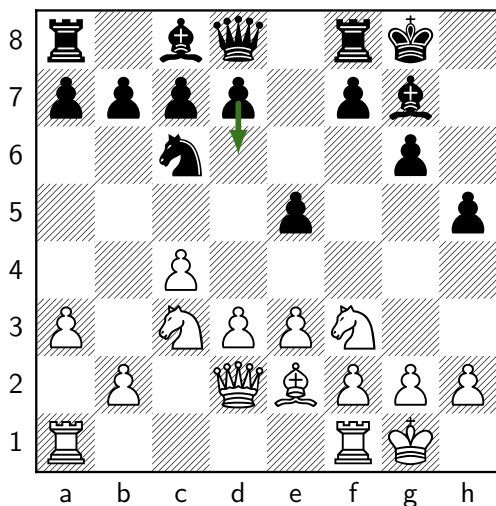
Figure 2: **Move 8, White: h4.** (a) Our position; Melanie plays **h4** (**h2h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1061` (ply 20) [structural distance $d=79$, QAP obj=26]: the master played **Bf1**; under μ our *Ph2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Nf3* as their *Bc1*, our *Bc3* as their *Pc3*; transferred move **h2h4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#1372` (ply 21) [structural distance $d=58$, QAP obj=24]: the master played **d6**; under μ our *Ph2* plays the role of that game’s *Pd2*; cross-type roles: our *pe6* as their *Ra8*, our *Ra1* as their *Qd8*; transferred move **h2h4** (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#3278` (ply 21) [structural distance $d=61$, QAP obj=45]: the master played **c5**; under μ our *Pe2* plays the role of that game’s *Pc2*; cross-type roles: our *Nf3* as their *Bd7*, our *Bc3* as their *Pa6*; transferred move **e2e4**. Synthesis: 2 retrieved analogs transferred to **h4** (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



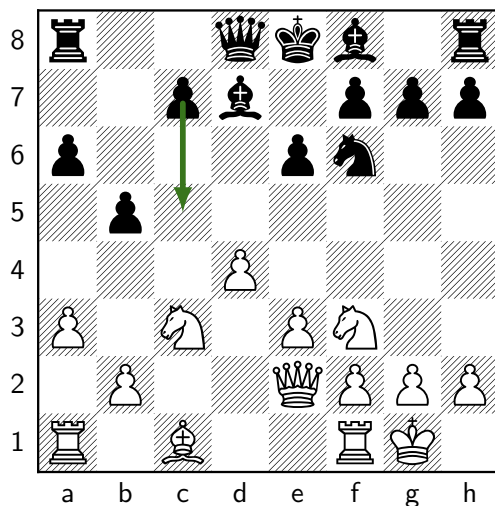
(a)



(b)

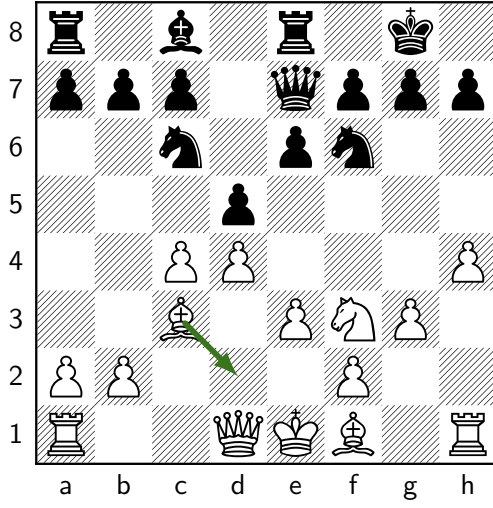


(c)

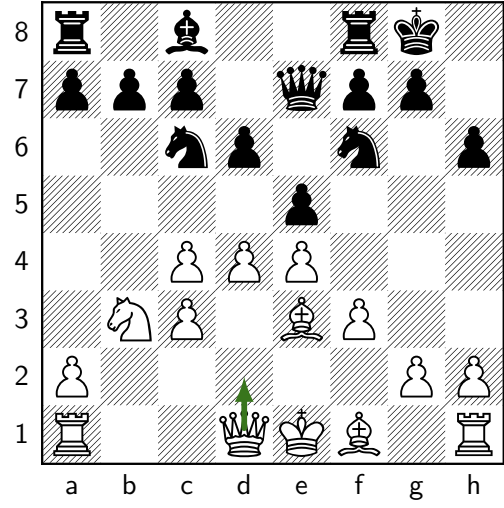


(d)

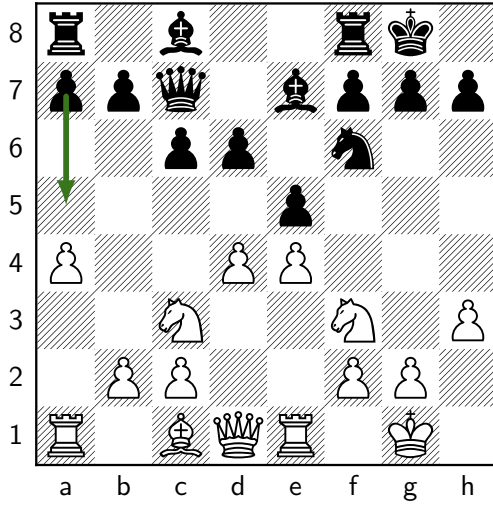
Figure 3: **Move 9, White: e3 (e2e3).** (a) Our position; Melanie plays **e3 (e2e3)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#319` (ply 21) [structural distance $d=78$, QAP obj=28]: the master played **d6**; under μ our *Pe2* plays the role of that game’s *Pd2*; cross-type roles: our *nf6* as their *Ra8*, our *Ra1* as their *Qd8*; transferred move **e2e3 (endorses this move)**. (c) Retrieved from `lichess_elite_2024-09_p0025#1372` (ply 21) [structural distance $d=62$, QAP obj=25]: the master played **d6**; under μ our *Pe2* plays the role of that game’s *Pd2*; cross-type roles: our *pe6* as their *Ra8*, our *Bc3* as their *Qd8*; transferred move **e2e3 (endorses this move)**. (d) Retrieved from `lichess_elite_2025-11_p0000#3278` (ply 21) [structural distance $d=73$, QAP obj=44]: the master played **c5**; under μ our *Pe2* plays the role of that game’s *Pc2*; cross-type roles: our *Pc4* as their *Bf8*, our *Nf3* as their *Bd7*; transferred move **e2e4**. Synthesis: 2 retrieved analogs transferred to **e3** (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



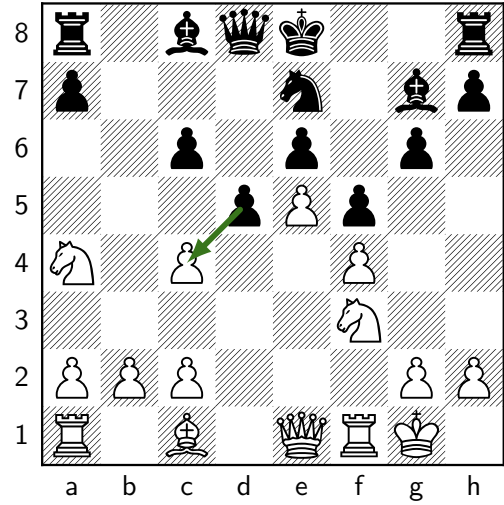
(a)



(b)

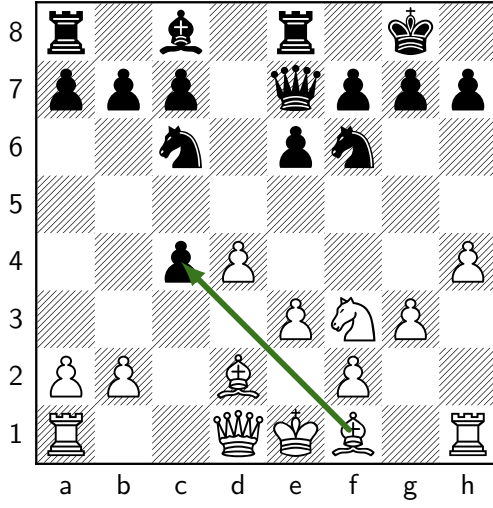


(c)

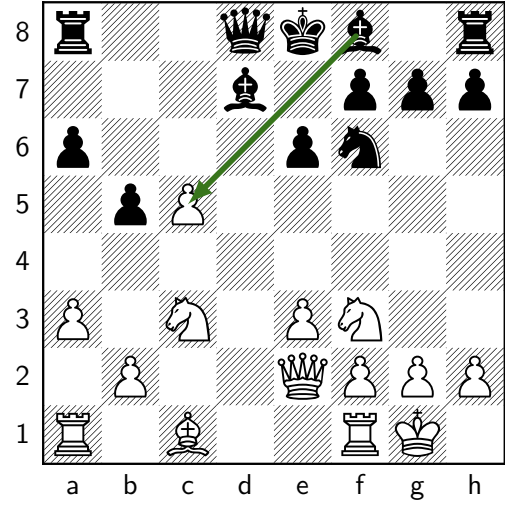


(d)

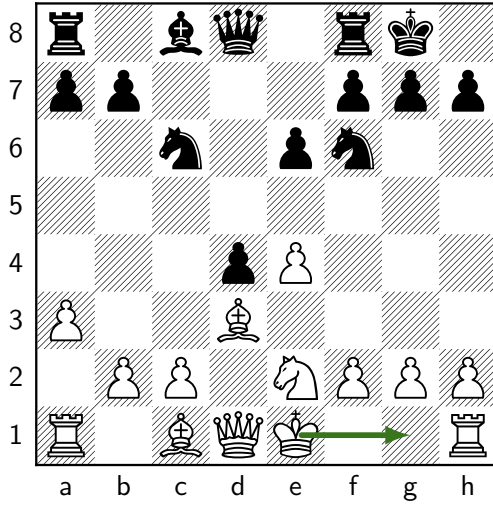
Figure 4: **Move 10, White: Bd2.** (a) Our position; Melanie plays Bd2 (c3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#2227` (ply 20) [structural distance $d=69$, QAP obj=61]: the master played Qd2; under μ our *Bc3* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Pa2* as their *Ra1*, our *Ra1* as their *Pa2*; transferred move c3d2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#1603` (ply 21) [structural distance $d=76$, QAP obj=60]: the master played a5; under μ our *Ra1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Ra8*, our *nf6* as their *Bc8*; transferred move a1b1. (d) Retrieved from `lichess_elite_2024-09_p0023#1924` (ply 23) [structural distance $d=76$, QAP obj=54]: the master played dxc4; under μ our *Pc4* plays the role of that game’s *Pd4*; cross-type roles: our *qe7* as their *Ra8*, our *Nf3* as their *Qd8*; transferred move c4c5. Synthesis: 1 retrieved analog transferred to Bd2 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



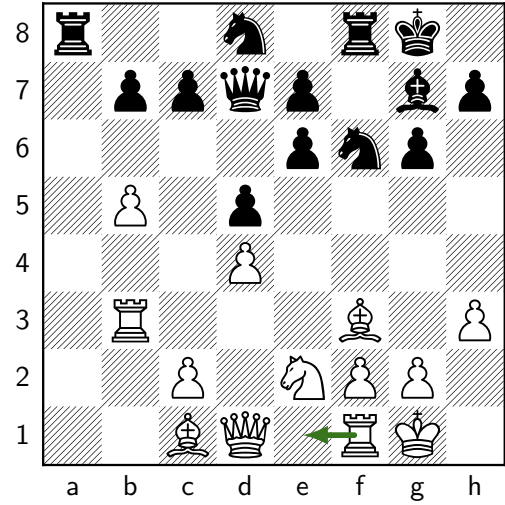
(a)



(b)

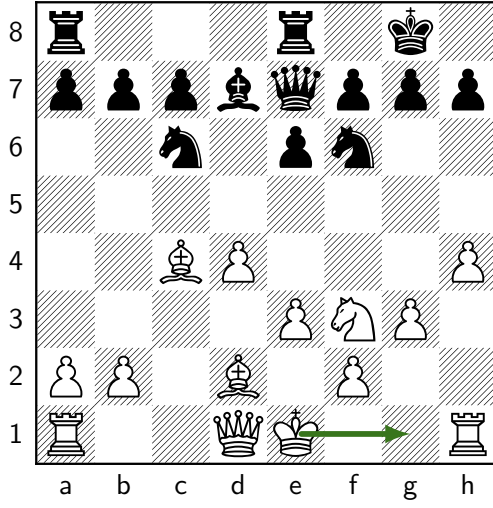


(c)

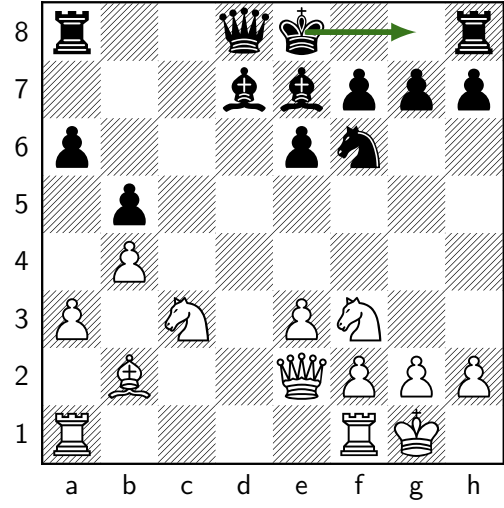


(d)

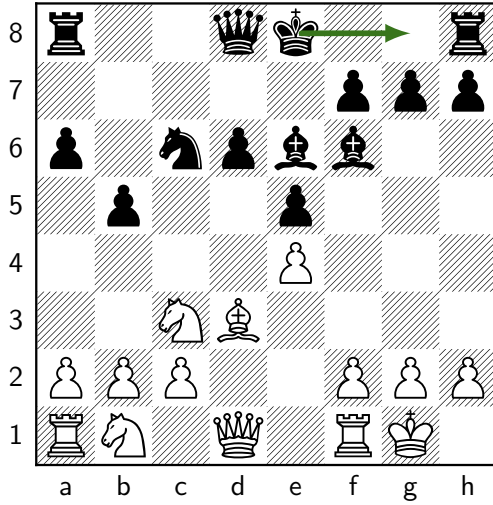
Figure 5: **Move 11, White: Bxc4.** (a) Our position; Melanie plays Bxc4 (f1c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#3278` (ply 23) [structural distance $d=46$, QAP obj=61]: the master played Bxc5; under μ our *Bf1* plays the role of that game’s *Bf1*; cross-type roles: our *qe7* as their *pa3*, our *nc6* as their *qe2*; transferred move f1c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#515` (ply 20) [structural distance $d=77$, QAP obj=40]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pa2* as their *Bc1*, our *Bd2* as their *Rh1*; transferred move e1e2. (d) Retrieved from `lichess_elite_2024-09_p0031#2850` (ply 34) [structural distance $d=73$, QAP obj=40]: the master played Re1; under μ our *Qd1* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *pe6* as their *Bc1*, our *ra8* as their *Qd1*; transferred move d1b1. Synthesis: 1 retrieved analog transferred to Bxc4 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



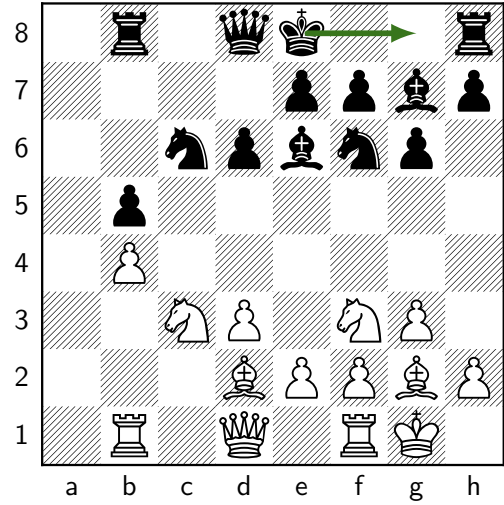
(a)



(b)

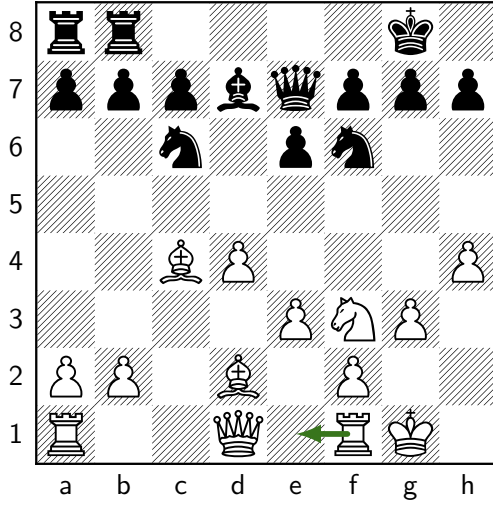


(c)

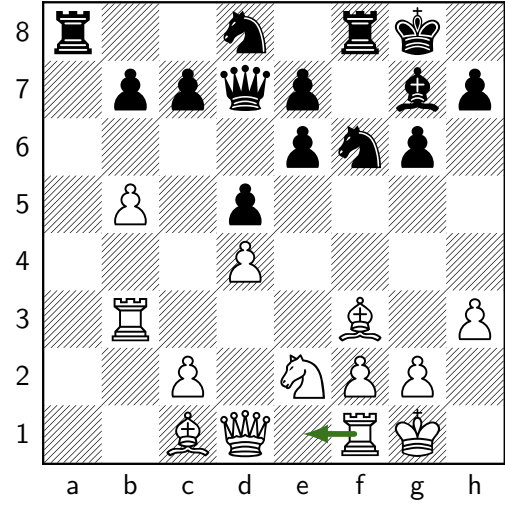


(d)

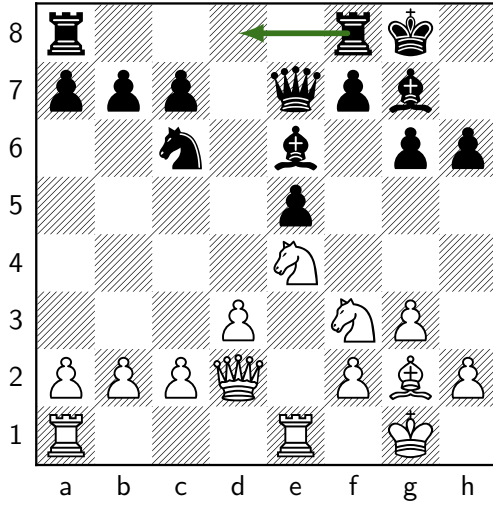
Figure 6: **Move 12, White: 0-0.** (a) Our position; Melanie plays 0-0 (**e1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#3278` (ply 27) [structural distance $d=77$, QAP obj=58]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *qe7* as their *pe3*, our *nc6* as their *qe2*; transferred move **e1g1** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#945` (ply 23) [structural distance $d=73$, QAP obj=52]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bd2* as their *Qd8*, our *Pa2* as their *Rh8*; transferred move **e1g1** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#3772` (ply 25) [structural distance $d=73$, QAP obj=49]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *qe7* as their *Rb8*, our *Bd2* as their *Pe7*; transferred move **e1g1** (**endorses this move**). Synthesis: 7 retrieved analogs transferred to 0-0 (summed QAP objective 282, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



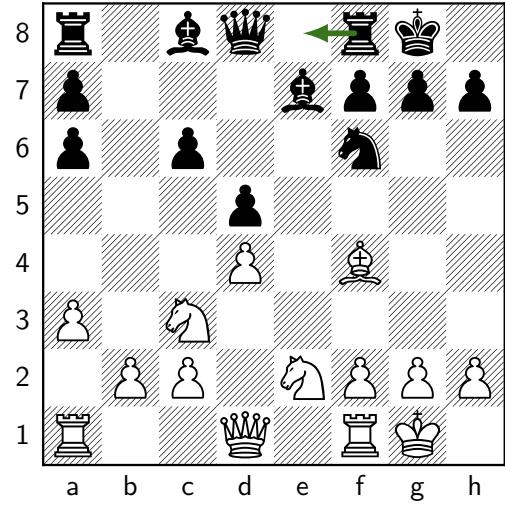
(a)



(b)

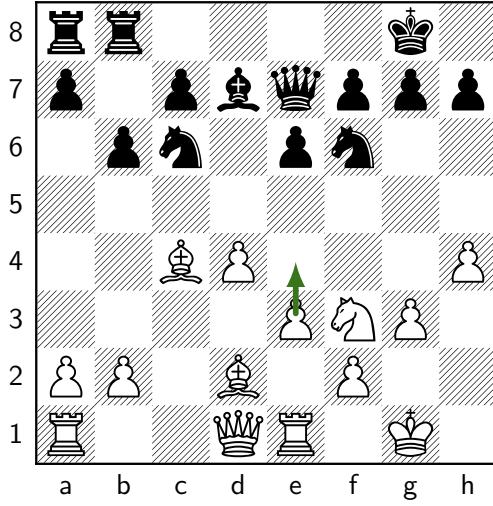


(c)

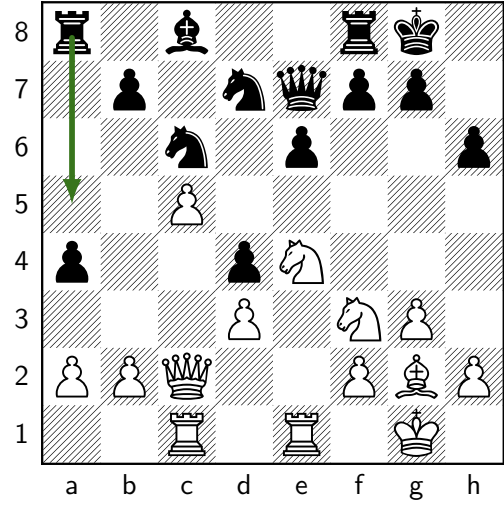


(d)

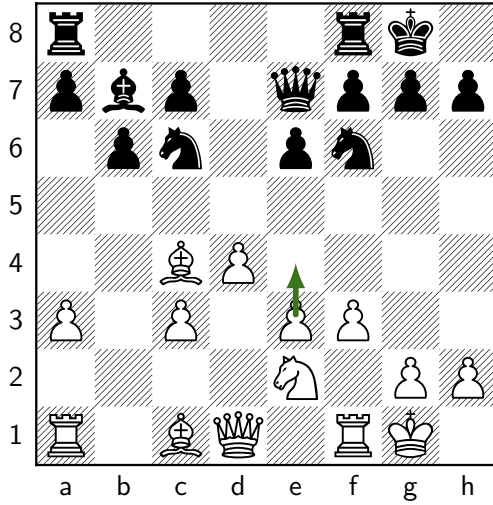
Figure 7: **Move 13, White: Re1.** (a) Our position; Melanie plays Re1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0031#2850 (ply 34) [structural distance $d=64$, QAP obj=47]: the master played Re1; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our ra8 as their Bc1, our nf6 as their Qd1; transferred move f1e1 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0018#1375 (ply 25) [structural distance $d=59$, QAP obj=38]: the master played Rfd8; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Qd1 as their Pa7, our nf6 as their Qe7; transferred move f1e1 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0028#724 (ply 23) [structural distance $d=57$, QAP obj=35]: the master played Re8; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pe3 as their Ra8, our Ph4 as their Bc8; transferred move f1e1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re1 (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



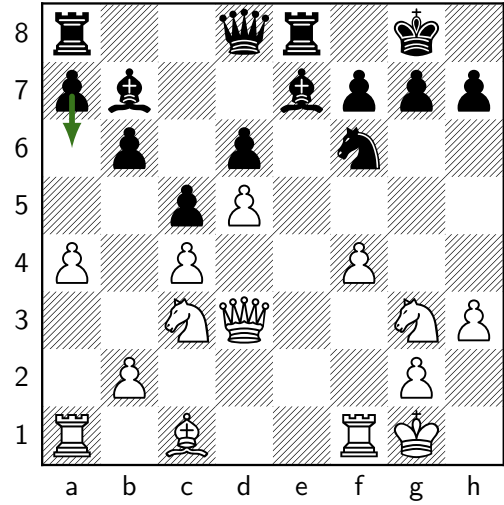
(a)



(b)

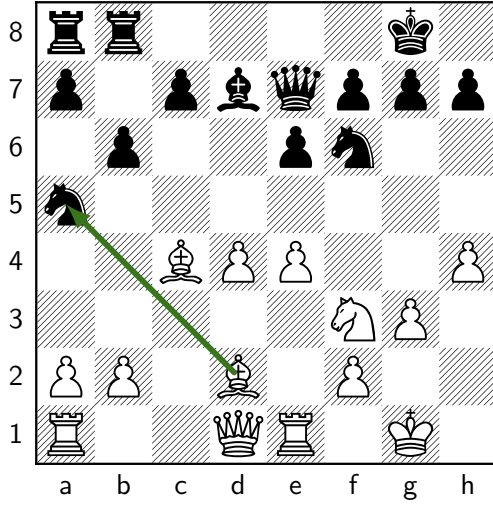


(c)

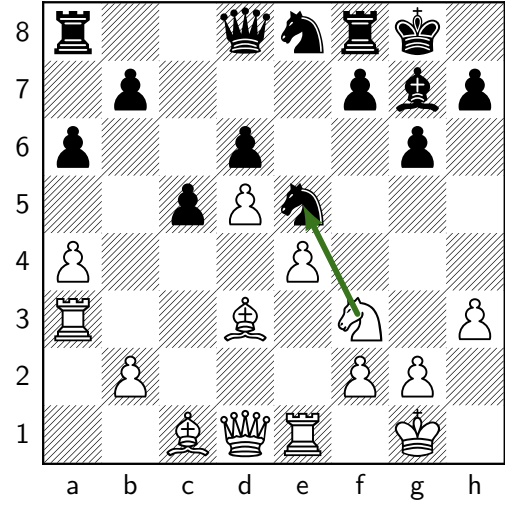


(d)

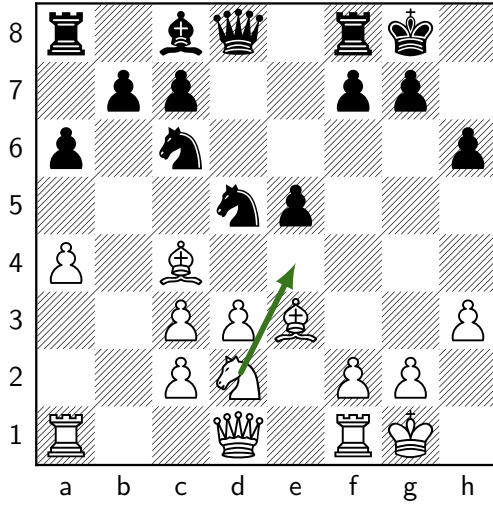
Figure 8: **Move 14, White: e4.** (a) Our position; Melanie plays e4 (e3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0016#3801 (ply 31) [structural distance $d=79$, QAP obj=65]: the master played Ra5; under μ our *Pe3* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Bc8*, our *Pa2* as their *Rf8*; transferred move e3e4 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0021#831 (ply 22) [structural distance $d=68$, QAP obj=47]: the master played e4; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *pb6* as their *Ra1*, our *Nf3* as their *Rf1*; transferred move e3e4 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0026#2025 (ply 31) [structural distance $d=78$, QAP obj=49]: the master played a6; under μ our *Qd1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *pa7* as their *Ra8*, our *ra8* as their *Qd8*; transferred move d1c2. Synthesis: 2 retrieved analogs transferred to e4 (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



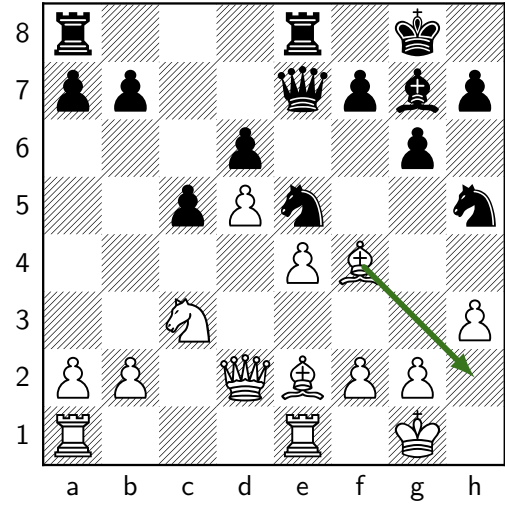
(a)



(b)

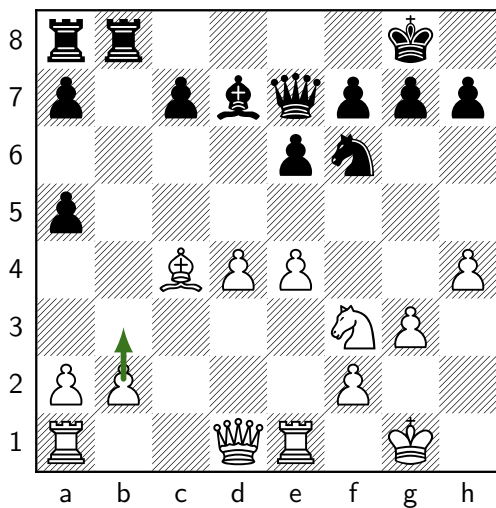


(c)

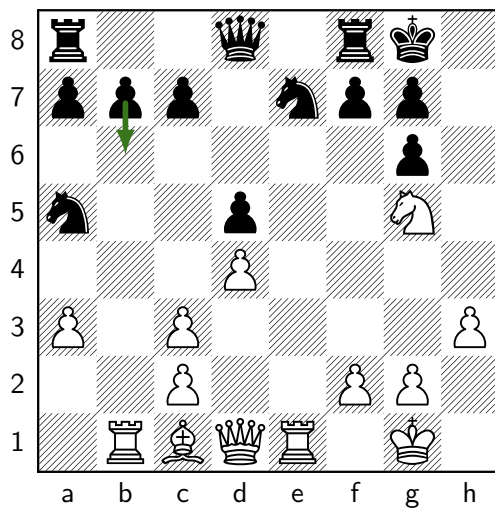


(d)

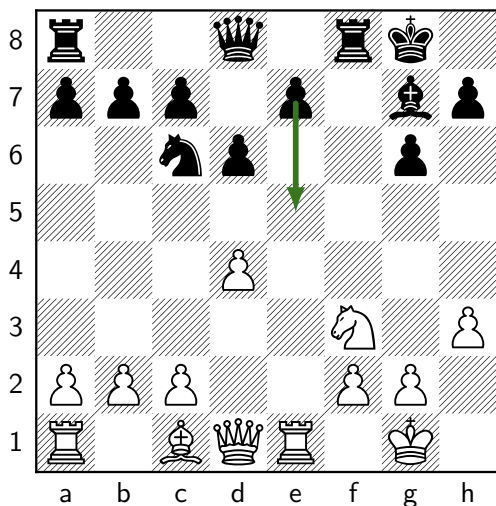
Figure 9: **Move 15, White: Bxa5.** (a) Our position; Melanie plays Bxa5 (d2a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#715` (ply 28) [structural distance $d=75$, QAP obj=51]: the master played Nxe5; under μ our Bd2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our pg7 as their Bc1, our Pd4 as their Re1; transferred move d2a5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#3894` (ply 24) [structural distance $d=78$, QAP obj=45]: the master played Ne4; under μ our Qd1 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our pe6 as their Ra1, our Pg3 as their Qd1; transferred move d1b1. (d) Retrieved from `lichess_elite_2024-09_p0028#3842` (ply 30) [structural distance $d=66$, QAP obj=44]: the master played Bh2; under μ our Bc4 plays the role of that game’s Bf4; cross-type roles: our Pe4 as their Ra1, our Nf3 as their Re1; transferred move c4f1. Synthesis: 1 retrieved analog transferred to Bxa5 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



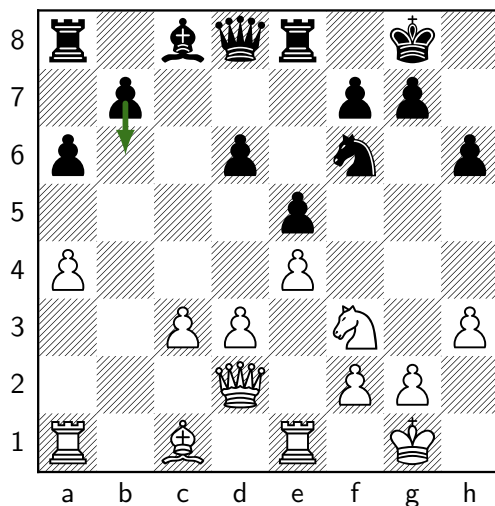
(a)



(b)

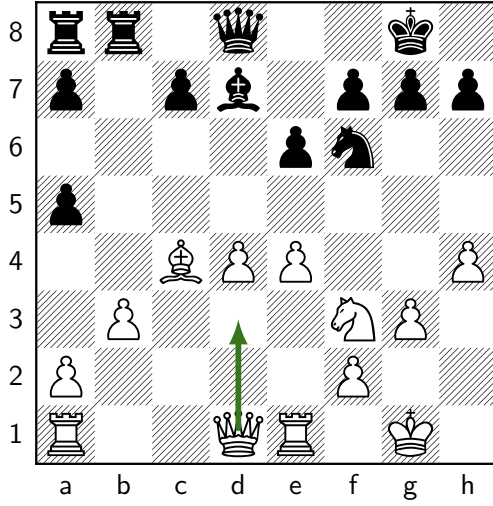


(c)

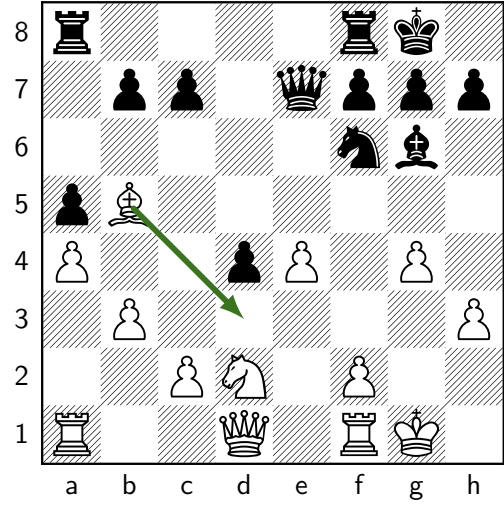


(d)

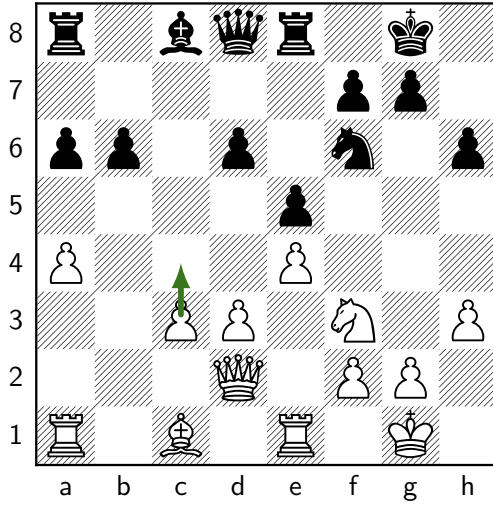
Figure 10: **Move 16, White: b3.** (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#3778` (ply 27) [structural distance $d=63$, QAP obj=48]: the master played b6; under μ our $Pb2$ plays the role of that game’s $Pb2$; cross-type roles: our $Re1$ as their $Qd8$, our $Nf3$ as their $Rf8$; transferred move b2b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#2910` (ply 23) [structural distance $d=55$, QAP obj=47]: the master played e5; under μ our $Pb2$ plays the role of that game’s $Pe2$; cross-type roles: our $Pd4$ as their $Qd8$, our $Bc4$ as their $Rf8$; transferred move b2b3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#3212` (ply 27) [structural distance $d=72$, QAP obj=55]: the master played b6; under μ our $Pa2$ plays the role of that game’s $Pb2$; cross-type roles: our $Pf2$ as their $Ra8$, our $qe7$ as their $Pa6$; transferred move a2a4. Synthesis: 2 retrieved analogs transferred to b3 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



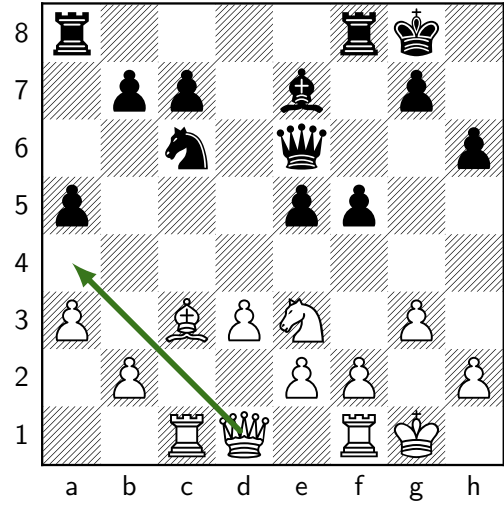
(a)



(b)

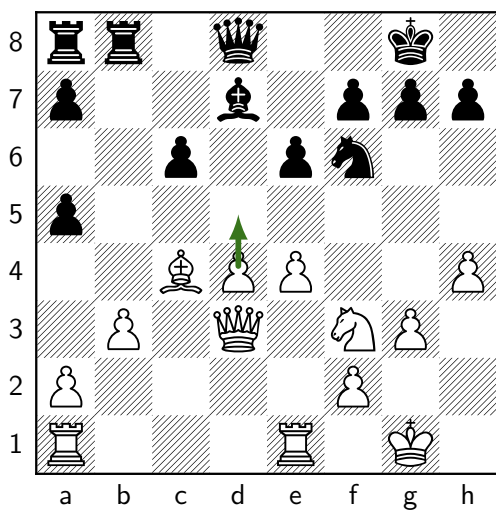


(c)

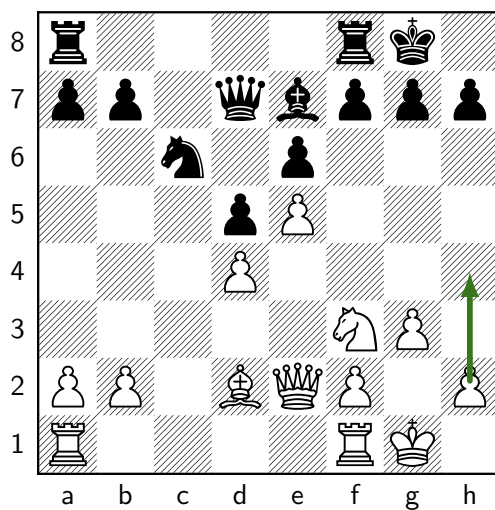


(d)

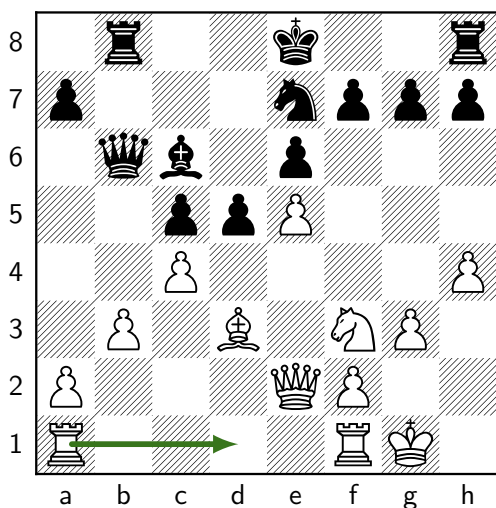
Figure 11: **Move 17, White: Qd3.** (a) Our position; Melanie plays Qd3 (d1d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#94` (ply 30) [structural distance $d=67$, QAP obj=72]: the master played Bd3; under μ our Qd1 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our Bc4 as their Ra1, our Pg3 as their Qd1; transferred move d1d3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#3212` (ply 28) [structural distance $d=64$, QAP obj=56]: the master played c4; under μ our Qd1 plays the role of that game’s Pc3 (a **cross-type** correspondence); cross-type roles: our pa5 as their Ra1, our pc7 as their Bc1; transferred move d1d3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#1806` (ply 34) [structural distance $d=66$, QAP obj=65]: the master played Qa4; under μ our Pg3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pg7 as their Rc1, our Nf3 as their Rf1; transferred move g3g4. Synthesis: 2 retrieved analogs transferred to Qd3 (summed QAP objective 127, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



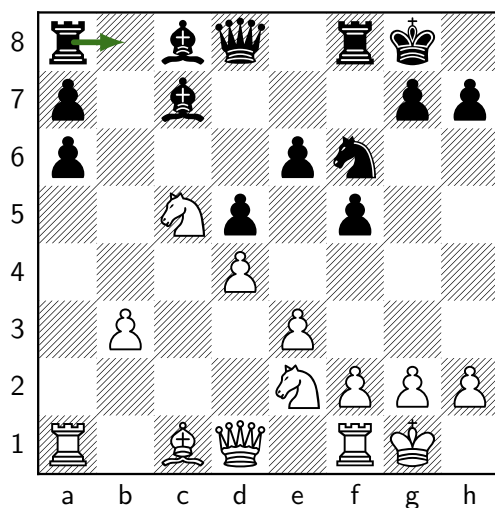
(a)



(b)

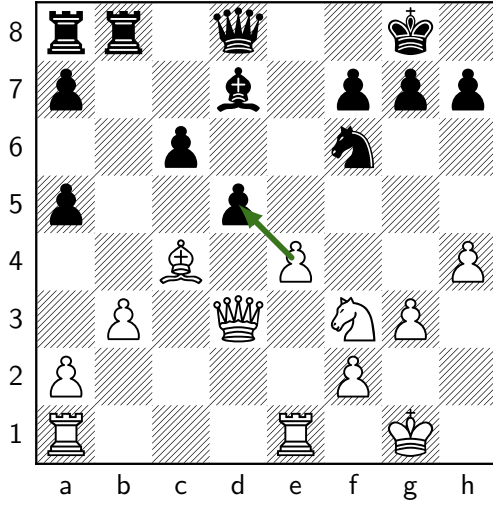


(c)

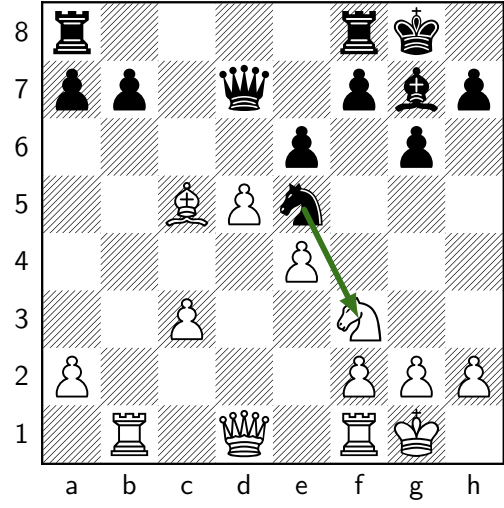


(d)

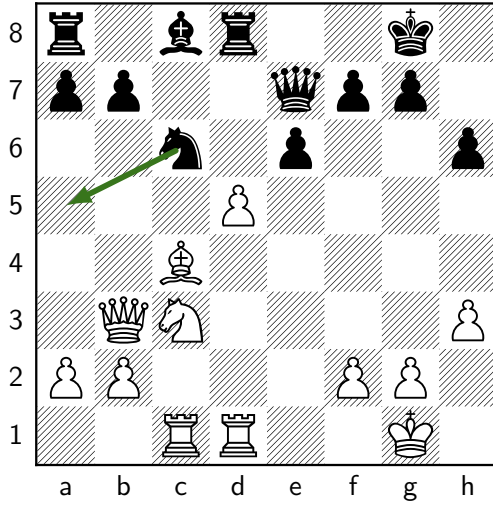
Figure 12: **Move 18, White: d5.** (a) Our position; Melanie plays d5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2097` (ply 26) [structural distance $d=71$, QAP obj=41]: the master played h4; under μ our $Pd4$ plays the role of that game’s $Ph2$; cross-type roles: our $Ph4$ as their $Qe2$, our $bd7$ as their $pa7$; transferred move d4d5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#733` (ply 36) [structural distance $d=72$, QAP obj=26]: the master played Rad1; under μ our $Pd4$ plays the role of that game’s $Ra1$ (a **cross-type** correspondence); cross-type roles: our $ra8$ as their $Qe2$, our $pa5$ as their $Bd3$; transferred move d4d5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#3358` (ply 27) [structural distance $d=64$, QAP obj=54]: the master played Rb8; under μ our $Ph4$ plays the role of that game’s $Ra1$ (a **cross-type** correspondence); cross-type roles: our $Qd3$ as their $Bc8$, our $ra8$ as their $Qd8$; transferred move h4h5. Synthesis: 2 retrieved analogs transferred to d5 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



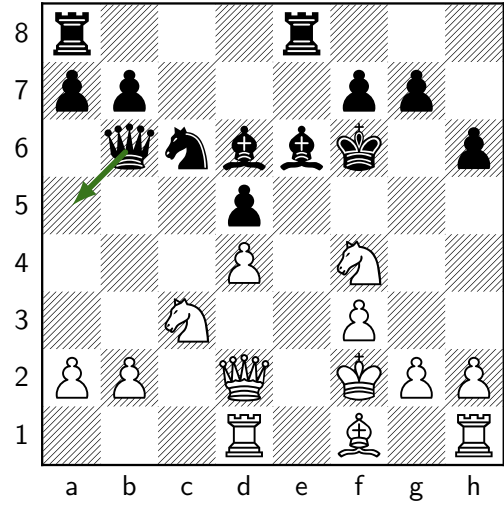
(a)



(b)

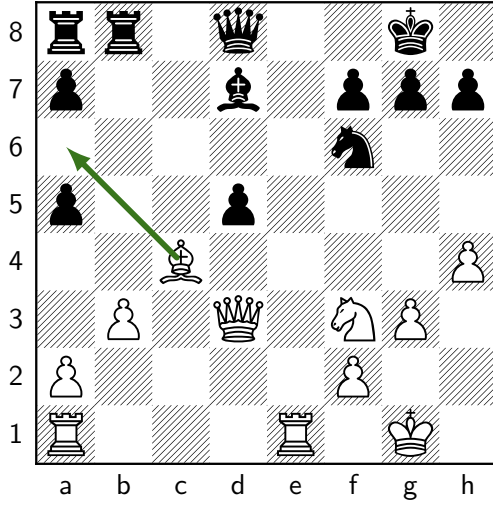


(c)

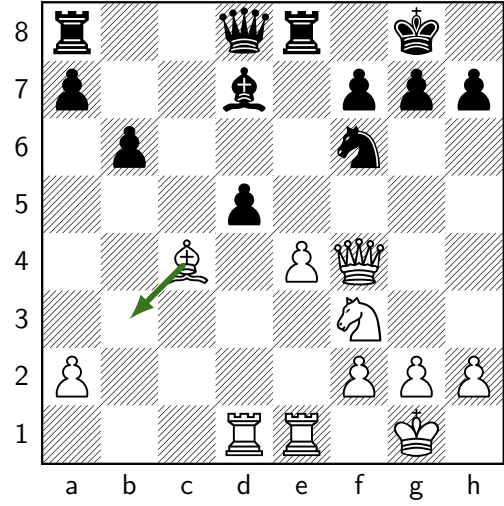


(d)

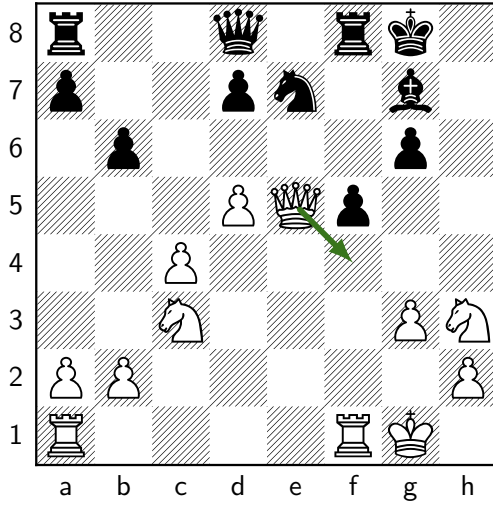
Figure 13: **Move 19, White: exd5.** (a) Our position; Melanie plays exd5 (e4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#1648` (ply 27) [structural distance $d=78$, QAP obj=90]: the master played $Nxf3+$; under μ our $Bc4$ plays the role of that game’s $Ne4$ (a **cross-type** correspondence); cross-type roles: our $Nf3$ as their $Pb7$, our $pa5$ as their $Bg7$; transferred move $c4b5$. (c) Retrieved from `lichess_elite_2024-09_p0031#2822` (ply 31) [structural distance $d=71$, QAP obj=88]: the master played $Na5$; under μ our $Bc4$ plays the role of that game’s $Nc3$ (a **cross-type** correspondence); cross-type roles: our $Re1$ as their $Bc8$, our $Qd3$ as their $Rd8$; transferred move $c4a6$. (d) Retrieved from `lichess_elite_2024-09_p0012#3630` (ply 31) [structural distance $d=77$, QAP obj=76]: the master played $Qa5$; under μ our $Qd3$ plays the role of that game’s $Qb3$; cross-type roles: our $pg7$ as their $Re8$, our $Nf3$ as their $Pb7$; transferred move $d3d2$. Synthesis: no retrieved analog transferred to exd5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



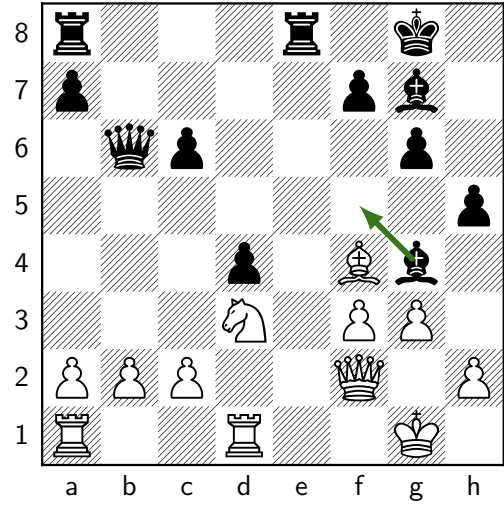
(a)



(b)

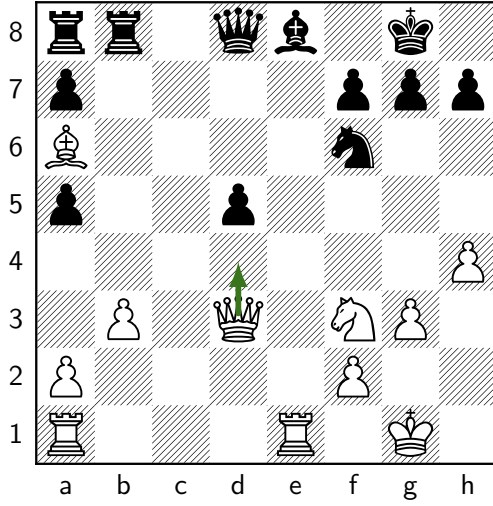


(c)

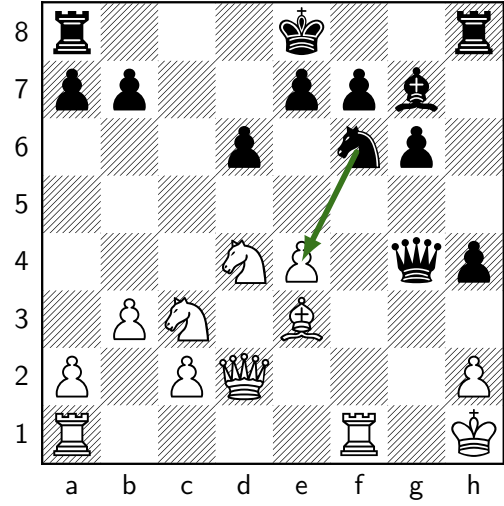


(d)

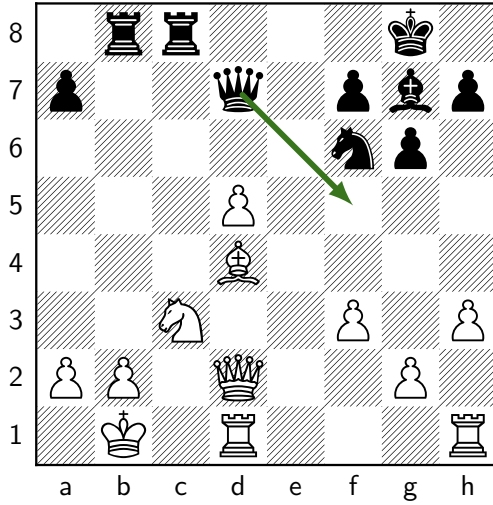
Figure 14: **Move 20, White: Ba6.** (a) Our position; Melanie plays Ba6 (c4a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#3128` (ply 32) [structural distance $d=78$, QAP obj=70]: the master played Bb3; under μ our Bc4 plays the role of that game’s Bc4; cross-type roles: our Qd3 as their Rd1, our Bc4 as their Pe4; transferred move c4a6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#980` (ply 36) [structural distance $d=80$, QAP obj=36]: the master played Qf4; under μ our Bc4 plays the role of that game’s Qe5 (a **cross-type** correspondence); cross-type roles: our Qd3 as their Rf1, our Re1 as their Pa2; transferred move c4a6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#3797` (ply 43) [structural distance $d=77$, QAP obj=30]: the master played Bf5; under μ our Bc4 plays the role of that game’s Bg5; cross-type roles: our Pa2 as their Re8, our ra8 as their Bg7; transferred move c4a6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ba6 (summed QAP objective 164, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



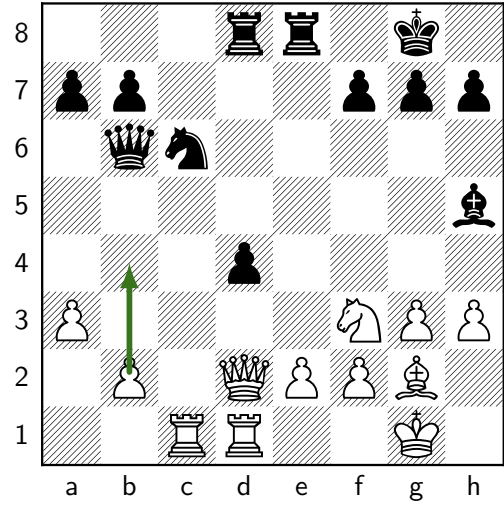
(a)



(b)

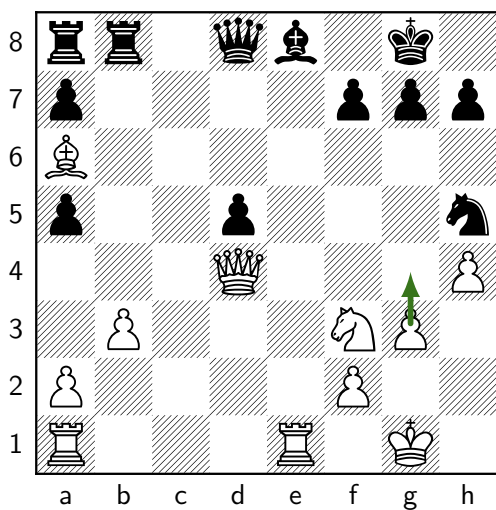


(c)

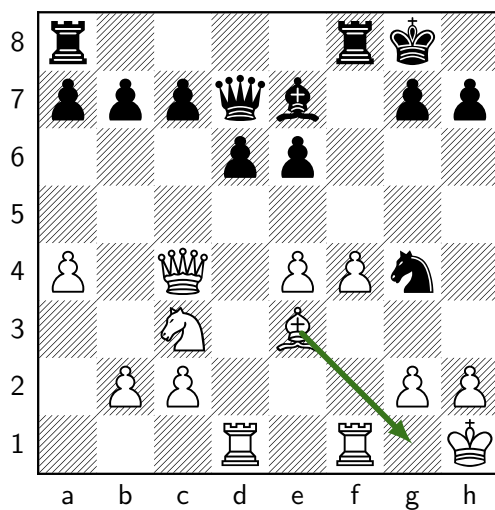


(d)

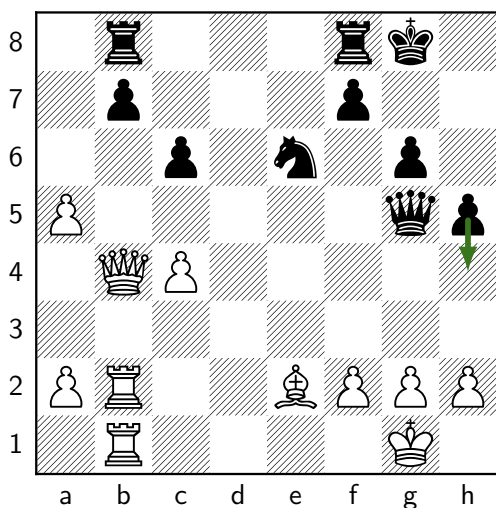
Figure 15: **Move 21, White: Qd4.** (a) Our position; Melanie plays Qd4 (d3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#2962` (ply 33) [structural distance $d=75$, QAP obj=54]: the master played Nxe4; under μ our Qd3 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Ra8, our Ba6 as their Pe7; transferred move d3d4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#1790` (ply 37) [structural distance $d=71$, QAP obj=50]: the master played Qf5+; under μ our Qd3 plays the role of that game’s Qd2; cross-type roles: our Nf3 as their Rb8, our Ba6 as their Pa7; transferred move d3f5. (d) Retrieved from `lichess_elite_2024-09_p0030#1206` (ply 36) [structural distance $d=59$, QAP obj=45]: the master played b4; under μ our Ra1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Rd1, our pa5 as their Bg2; transferred move a1b1. Synthesis: 1 retrieved analog transferred to Qd4 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



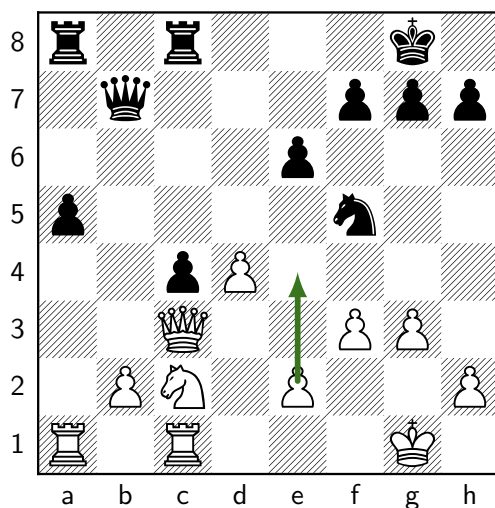
(a)



(b)

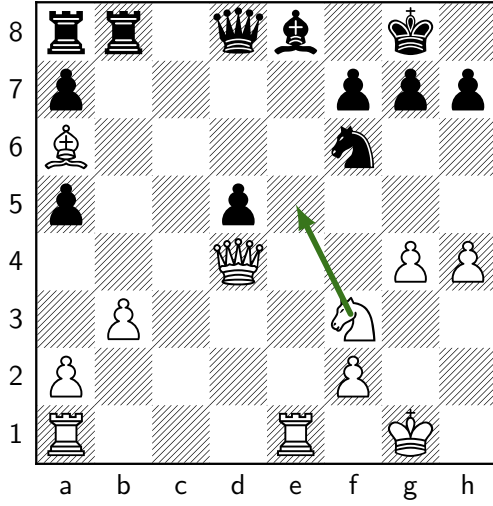


(c)

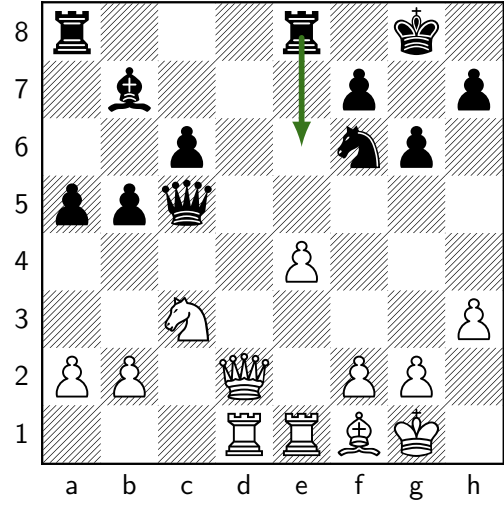


(d)

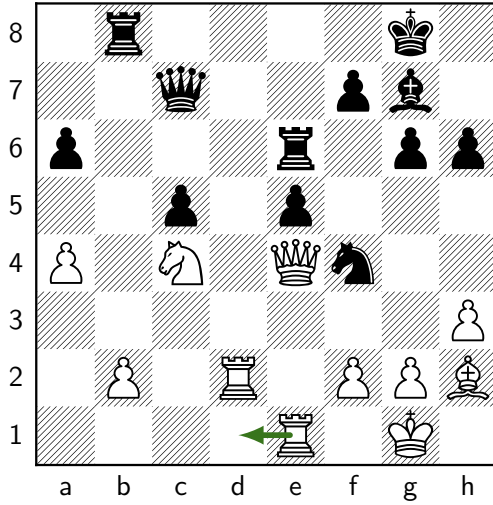
Figure 16: **Move 22, White: g4.** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#1102` (ply 34) [structural distance $d=77$, QAP obj=40]: the master played Bg1; under μ our $Pg3$ plays the role of that game’s $Be3$ (a **cross-type** correspondence); cross-type roles: our $ph7$ as their $Rd1$, our $Pf2$ as their $Rf1$; transferred move g3g4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#168` (ply 51) [structural distance $d=76$, QAP obj=39]: the master played h4; under μ our $Pg3$ plays the role of that game’s $Ph4$; cross-type roles: our $qd8$ as their $Rb8$, our $pd5$ as their $Rf8$; transferred move g3g4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#52` (ply 44) [structural distance $d=70$, QAP obj=78]: the master played e4; under μ our $Pb3$ plays the role of that game’s $Pe2$; cross-type roles: our $Qd4$ as their $Ra1$, our $Nf3$ as their $Rc1$; transferred move b3b4. Synthesis: 2 retrieved analogs transferred to g4 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



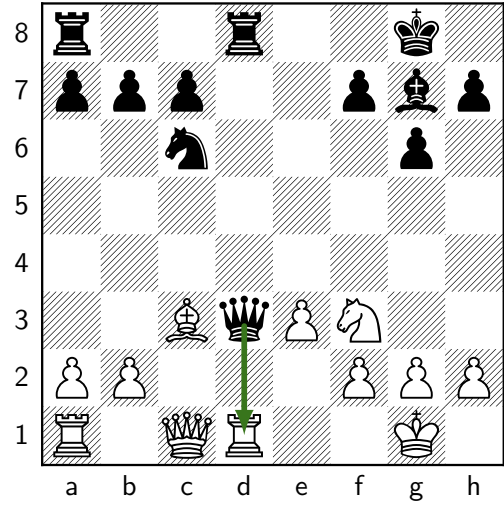
(a)



(b)

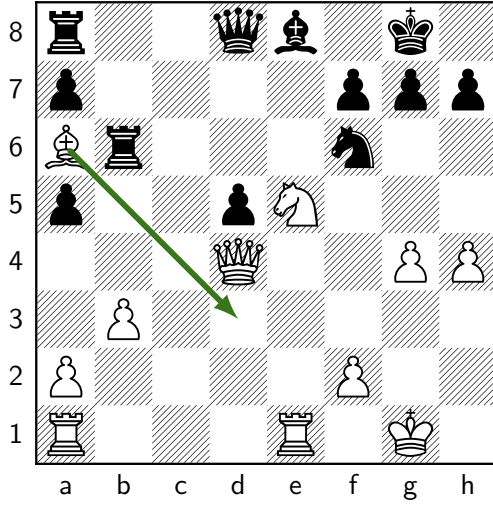


(c)

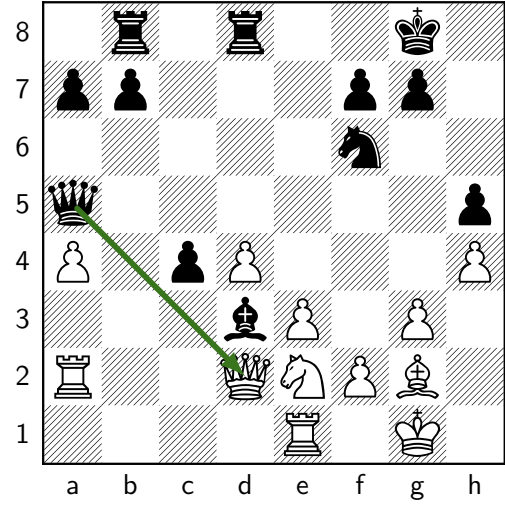


(d)

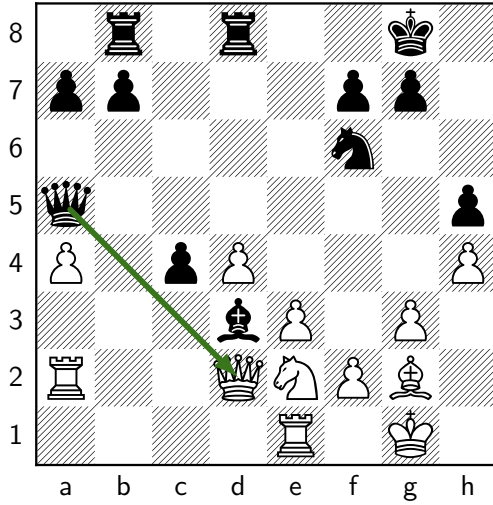
Figure 17: **Move 23, White: Ne5.** (a) Our position; Melanie plays Ne5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#1640` (ply 39) [structural distance $d=90$, QAP obj=68]: the master played Re6; under μ our Nf3 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our ph7 as their Ra8, our Ph4 as their Bb7; transferred move f3e5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#2471` (ply 58) [structural distance $d=85$, QAP obj=86]: the master played Red1; under μ our Nf3 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Pb3 as their Rd2, our Qd4 as their Bh2; transferred move f3d2. (d) Retrieved from `lichess_elite_2024-09_p0018#1195` (ply 33) [structural distance $d=80$, QAP obj=60]: the master played Qxd1+; under μ our Qd4 plays the role of that game’s Qd6; cross-type roles: our Nf3 as their Rd8, our Re1 as their Pa7; transferred move d4a7. Synthesis: 1 retrieved analog transferred to Ne5 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



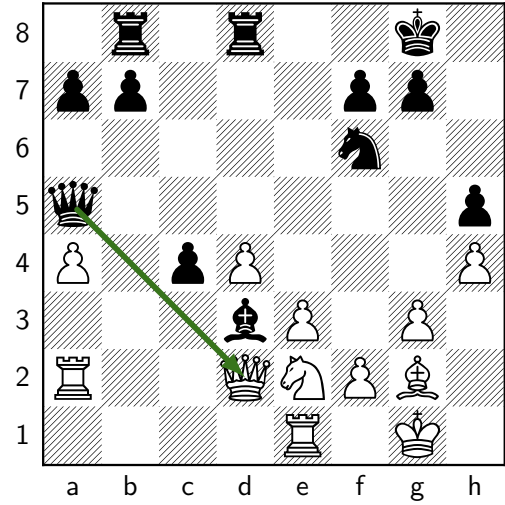
(a)



(b)

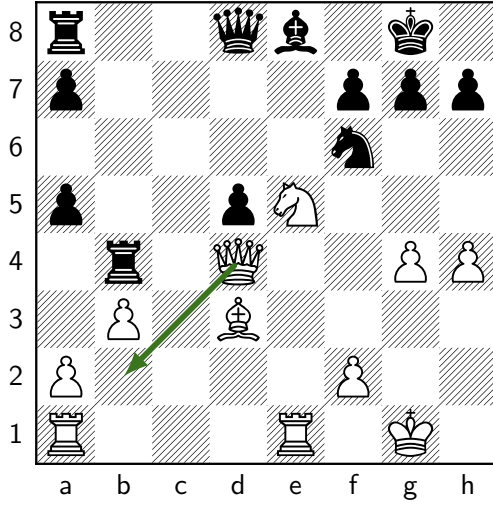


(c)

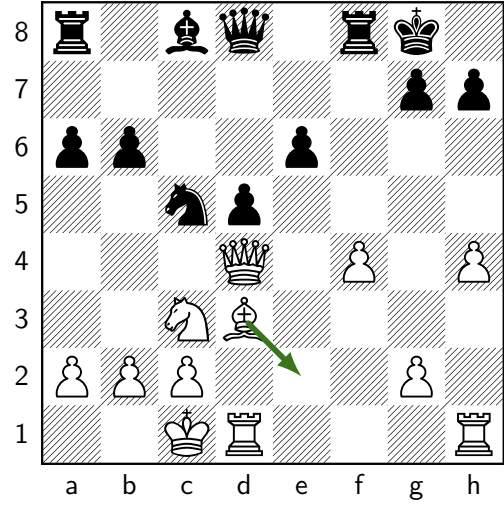


(d)

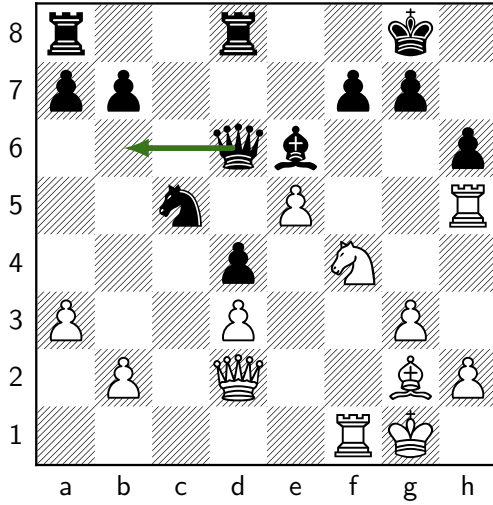
Figure 18: **Move 24, White: Bd3.** (a) Our position; Melanie plays Bd3 (a6d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2648` (ply 43) [structural distance $d=82$, QAP obj=80]: the master played Qxd2; under μ our Qd4 plays the role of that game’s Qa4; cross-type roles: our Pb3 as their Rb8, our Ne5 as their Rd8; transferred move d4b6. (c) Retrieved from `lichess_elite_2025-11_p0000#3035` (ply 43) [structural distance $d=82$, QAP obj=80]: the master played Qxd2; under μ our Qd4 plays the role of that game’s Qa4; cross-type roles: our Pb3 as their Rb8, our Ne5 as their Rd8; transferred move d4b6. (d) Retrieved from `lichess_elite_2025-11_p0000#3290` (ply 43) [structural distance $d=82$, QAP obj=80]: the master played Qxd2; under μ our Qd4 plays the role of that game’s Qa4; cross-type roles: our Pb3 as their Rb8, our Ne5 as their Rd8; transferred move d4b6. Synthesis: no retrieved analog transferred to Bd3 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



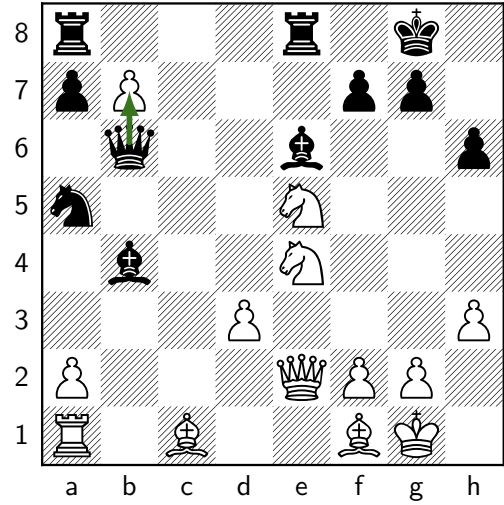
(a)



(b)

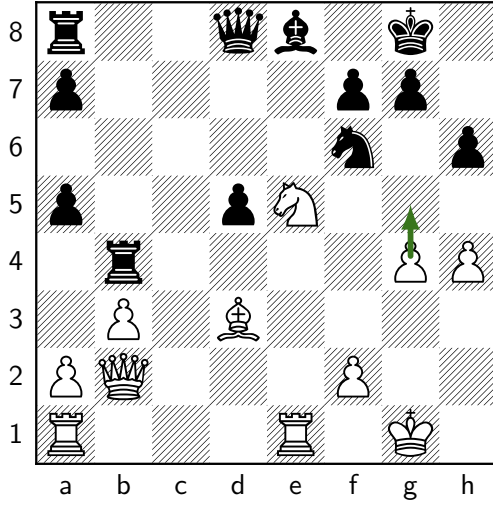


(c)

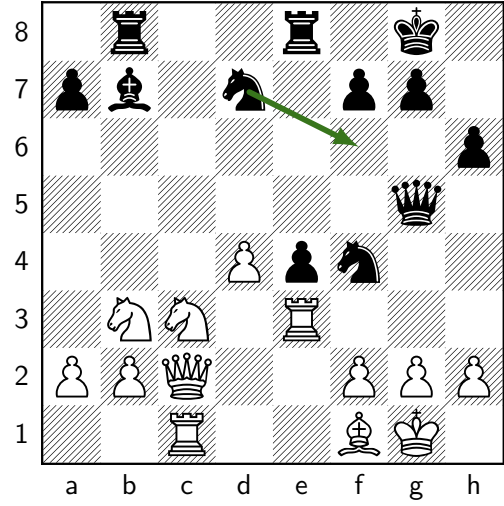


(d)

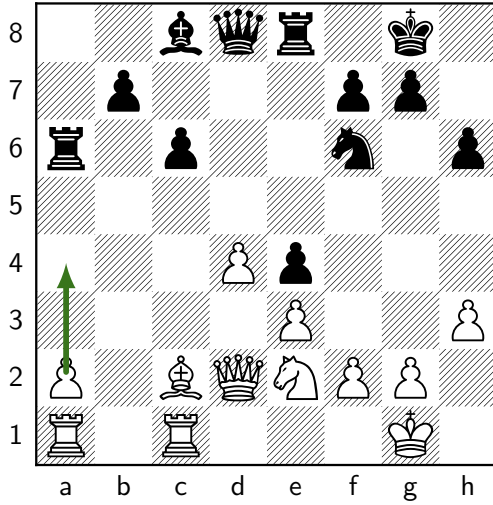
Figure 19: **Move 25, White: Qb2.** (a) Our position; Melanie plays Qb2 (d4b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2243` (ply 32) [structural distance $d=75$, QAP obj=60]: the master played Be2; under μ our Qd4 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our be8 as their Rd1, our Re1 as their Pa2; transferred move d4b2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#1777` (ply 43) [structural distance $d=86$, QAP obj=27]: the master played Qb6; under μ our Qd4 plays the role of that game’s Qd3; cross-type roles: our Pb3 as their Ra8, our Bd3 as their Rd8; transferred move d4b2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#598` (ply 45) [structural distance $d=105$, QAP obj=24]: the master played Qxb7; under μ our Qd4 plays the role of that game’s Qb3; cross-type roles: our Qd4 as their Ra8, our Ra1 as their Pa7; transferred move d4b2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qb2 (summed QAP objective 111, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



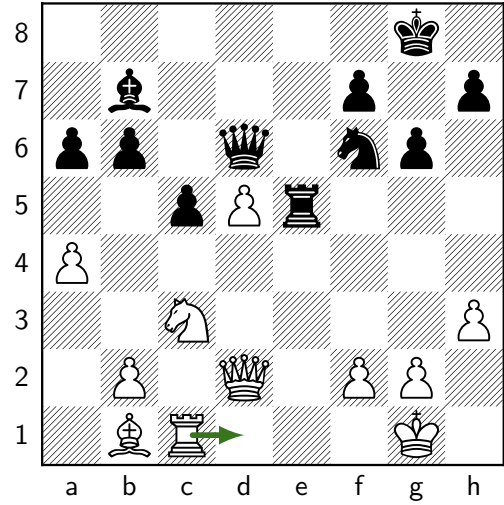
(a)



(b)

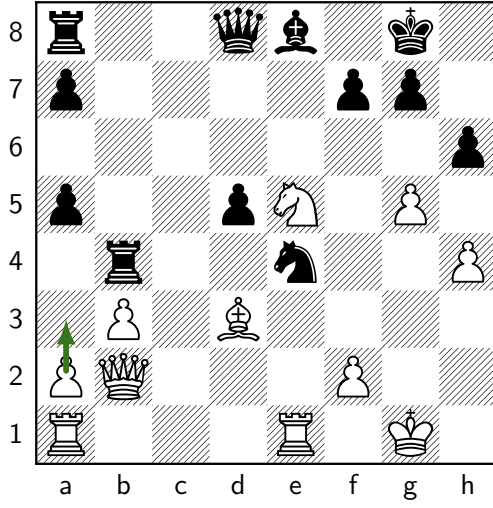


(c)

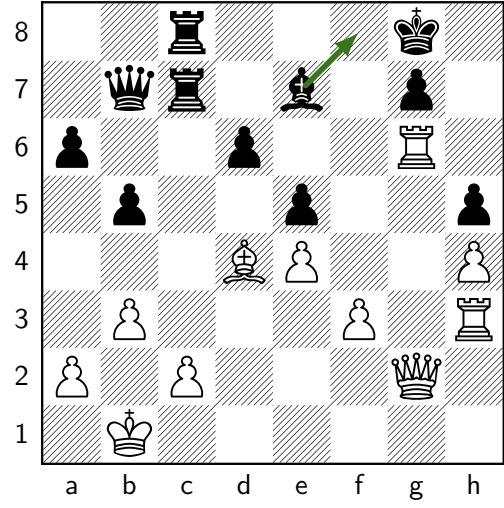


(d)

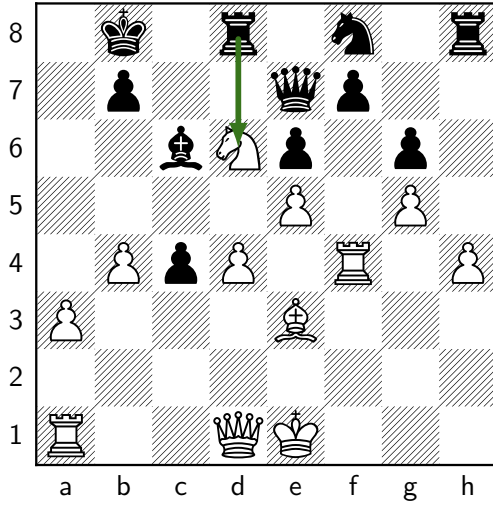
Figure 20: **Move 26, White: g5**. (a) Our position; Melanie plays g5 (g4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#3060` (ply 45) [structural distance $d=85$, QAP obj=65]: the master played Nf6; under μ our $Pg4$ plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Rb8, our Ra1 as their Bb7; transferred move g4g5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1617` (ply 40) [structural distance $d=80$, QAP obj=28]: the master played a4; under μ our $Pg4$ plays the role of that game’s Pa2; cross-type roles: our Qb2 as their Rc1, our Ne5 as their Bc2; transferred move g4g5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#3499` (ply 48) [structural distance $d=79$, QAP obj=70]: the master played Rd1; under μ our Re1 plays the role of that game’s Rc1; cross-type roles: our nf6 as their Ph3, our ra8 as their pc5; transferred move e1c1. Synthesis: 2 retrieved analogs transferred to g5 (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



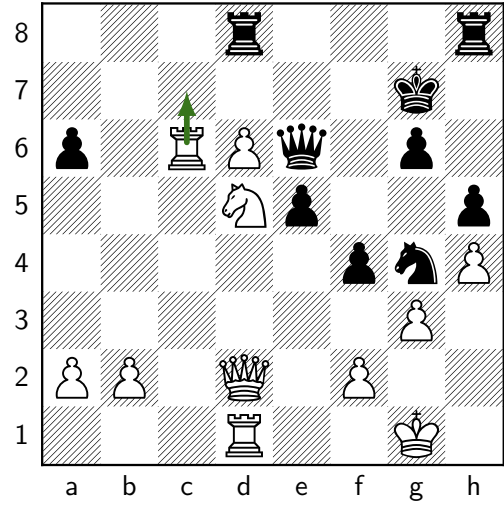
(a)



(b)

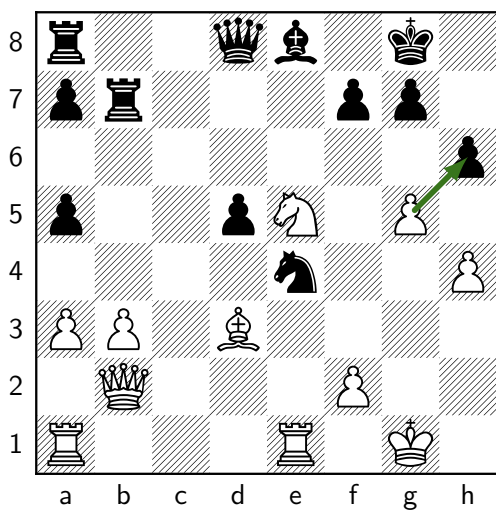


(c)

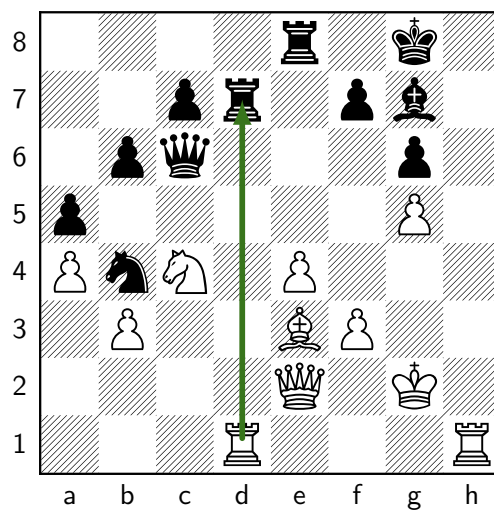


(d)

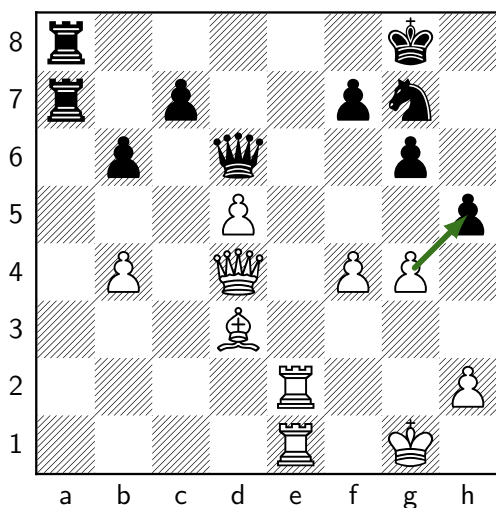
Figure 21: **Move 27, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#3339` (ply 55) [structural distance $d=77$, QAP obj=48]: the master played Bf8; under μ our Pa2 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our be8 as their Qb7, our ph6 as their Rc7; transferred move a2a3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#3949` (ply 51) [structural distance $d=82$, QAP obj=69]: the master played Rxd6; under μ our Re1 plays the role of that game’s Rd1; cross-type roles: our pa5 as their Nf8, our Ne5 as their Rh8; transferred move e1e4. (d) Retrieved from `lichess_elite_2024-09_p0013#2367` (ply 50) [structural distance $d=78$, QAP obj=67]: the master played Rc7+; under μ our Qb2 plays the role of that game’s Rc6 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pa2, our Pf2 as their ng4; transferred move b2c1. Synthesis: 1 retrieved analog transferred to a3 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



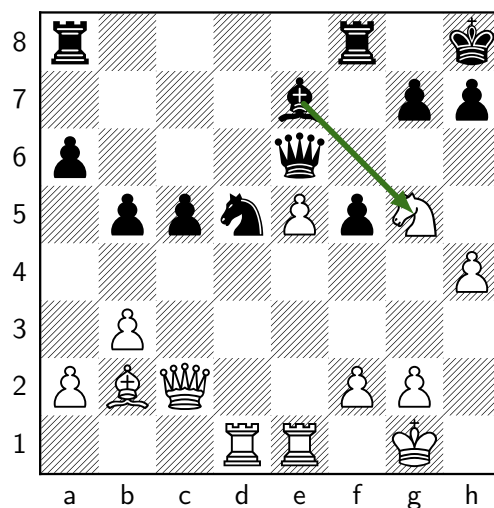
(a)



(b)

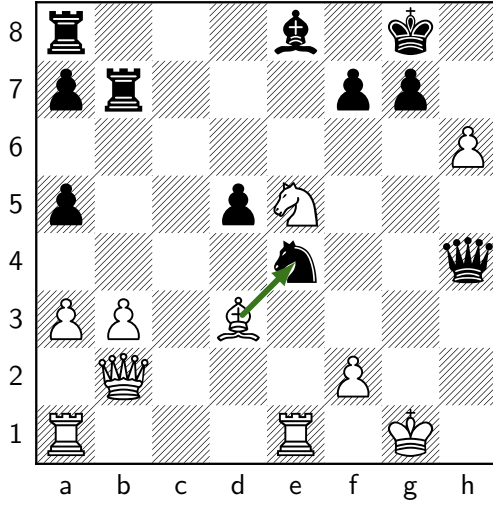


(c)

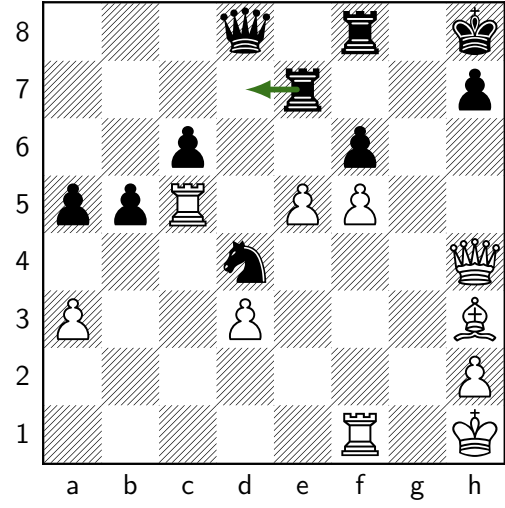


(d)

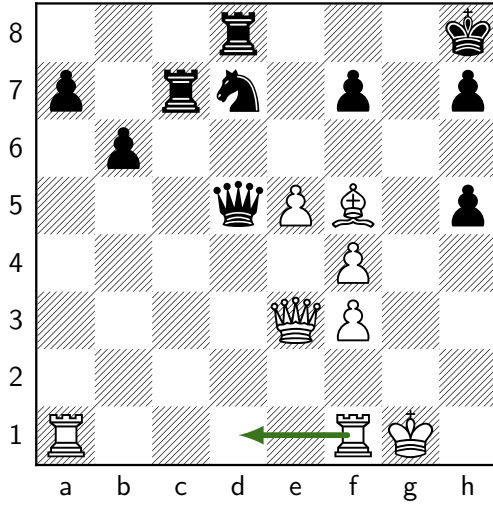
Figure 22: **Move 28, White: gxh6.** (a) Our position; Melanie plays gxh6 (g5h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0020#458 (ply 58) [structural distance $d=86$, QAP obj=47]: the master played Rxd7; under μ our Pg5 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Ne5 as their Rh1, our Re1 as their Qe2; transferred move g5h6 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0021#1492 (ply 68) [structural distance $d=86$, QAP obj=31]: the master played gxh5; under μ our Pg5 plays the role of that game’s Pg4; cross-type roles: our Pa3 as their Re1, our Qb2 as their Re2; transferred move g5h6 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0012#3847 (ply 47) [structural distance $d=88$, QAP obj=17]: the master played Bxg5; under μ our Pg5 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Ne5 as their Pg7, our ne4 as their Qe6; transferred move g5h6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to gxh6 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



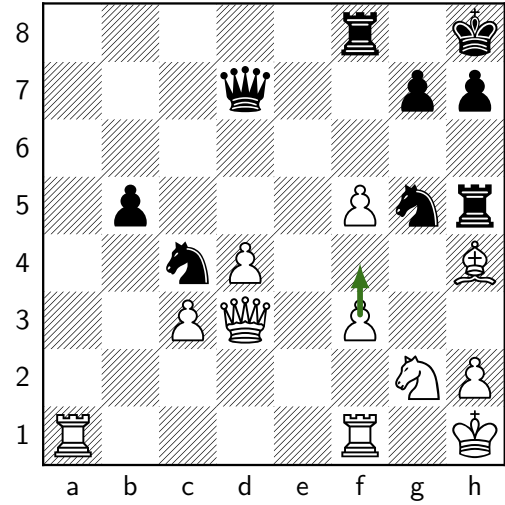
(a)



(b)

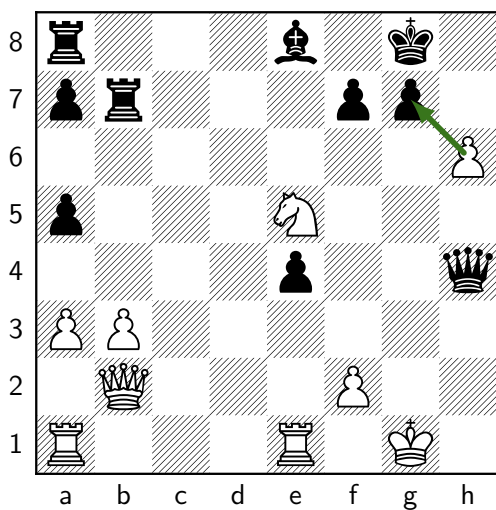


(c)

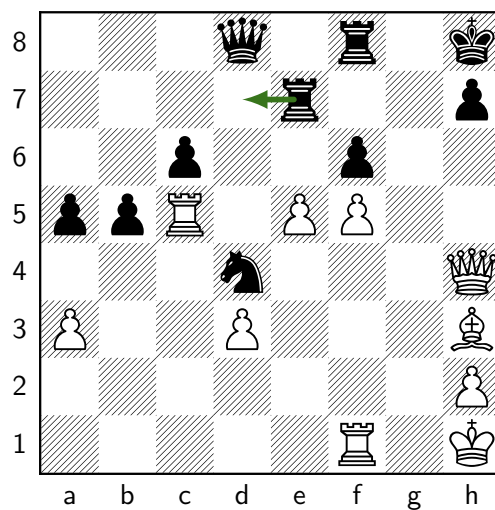


(d)

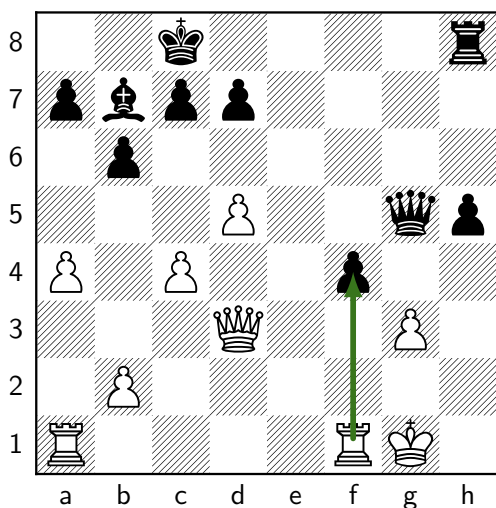
Figure 23: **Move 29, White: Bxe4.** (a) Our position; Melanie plays **Bxe4** (**d3e4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#604` (ply 59) [structural distance $d=69$, QAP obj=71]: the master played **Rd7**; under μ our *Re1* plays the role of that game’s *Re2*; cross-type roles: our *Ne5* as their *Qd8*, our *Qb2* as their *Rf8*; transferred move **e1e2**. (c) Retrieved from `lichess_elite_2024-09_p0012#376` (ply 52) [structural distance $d=84$, QAP obj=49]: the master played **Rfd1**; under μ our *Ra1* plays the role of that game’s *Rf1*; cross-type roles: our *Pa3* as their *Ra1*, our *Ne5* as their *Pf4*; transferred move **a1d1**. (d) Retrieved from `lichess_elite_2025-11_p0000#2644` (ply 54) [structural distance $d=86$, QAP obj=47]: the master played **f4**; under μ our *Ph6* plays the role of that game’s *Pf3*; cross-type roles: our *Qb2* as their *Ra1*, our *pd5* as their *Ng2*; transferred move **h6g7**. Synthesis: no retrieved analog transferred to **Bxe4** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



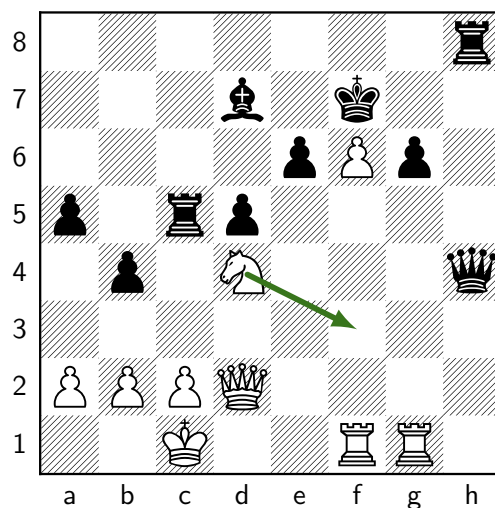
(a)



(b)

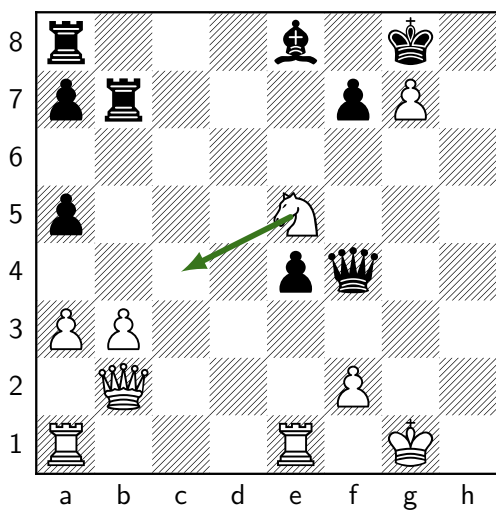


(c)

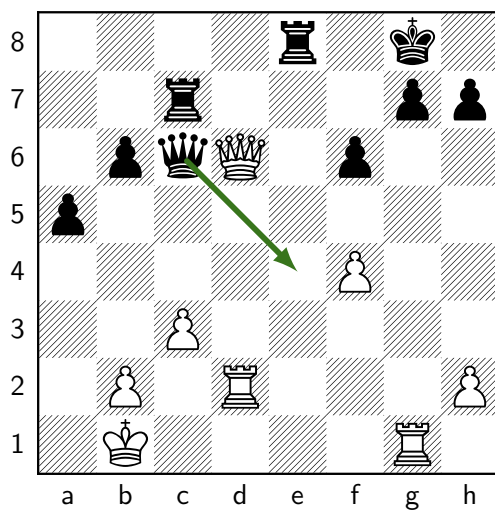


(d)

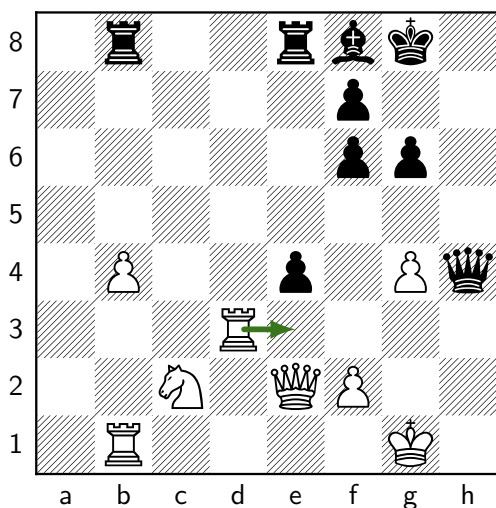
Figure 24: **Move 30, White: hxg7**. (a) Our position; Melanie plays **hxg7** (**h6g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#604` (ply 59) [structural distance $d=73$, QAP obj=127]: the master played **Rd7**; under μ our *Ph6* plays the role of that game’s *Re2* (a **cross-type** correspondence); cross-type roles: our *pg7* as their *rc5*, our *Ph6* as their *Nd4*; transferred move **h6g7** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#157` (ply 50) [structural distance $d=72$, QAP obj=70]: the master played **Rxf4**; under μ our *Ph6* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Ne5* as their *Qd3*, our *be8* as their *Pc4*; transferred move **h6g7** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#1004` (ply 56) [structural distance $d=71$, QAP obj=59]: the master played **Nf3**; under μ our *Ph6* plays the role of that game’s *Nd4* (a **cross-type** correspondence); cross-type roles: our *Ne5* as their *Rg1*, our *Qb2* as their *Pb2*; transferred move **h6g7** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **hxg7** (summed QAP objective 313, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



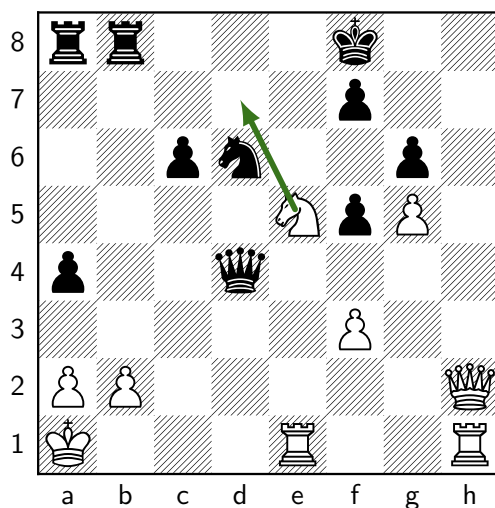
(a)



(b)

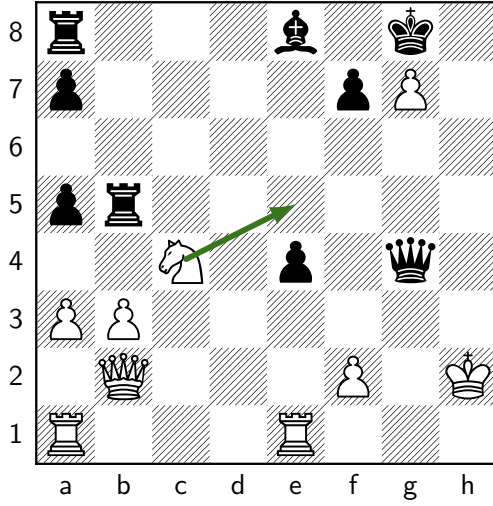


(c)

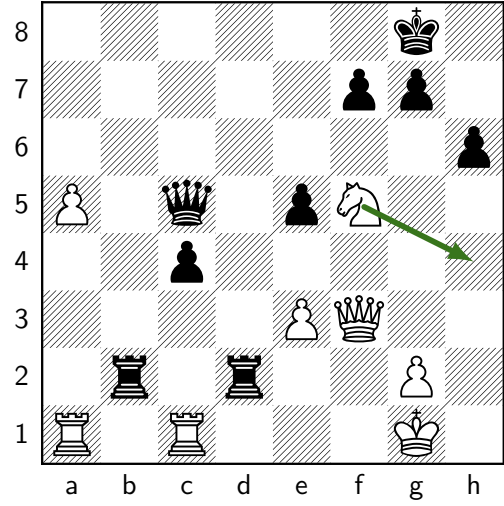


(d)

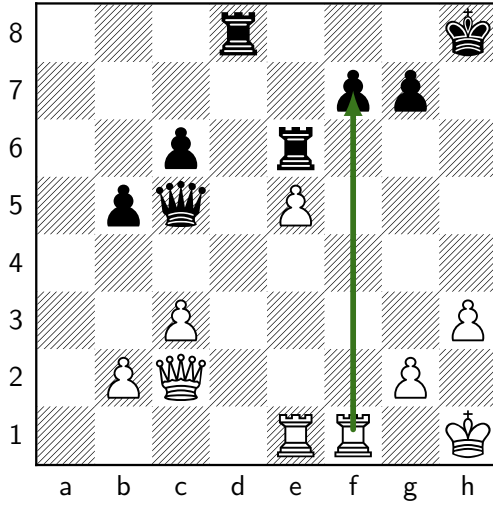
Figure 25: **Move 31, White: Nc4**. (a) Our position; Melanie plays Nc4 (e5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#2872` (ply 63) [structural distance $d=82$, QAP obj=26]: the master played Qe4+; under μ our Ne5 plays the role of that game’s Qc3 (a **cross-type** correspondence); cross-type roles: our Ne5 as their Rc7, our Qb2 as their Pf6; transferred move e5c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#243` (ply 62) [structural distance $d=86$, QAP obj=24]: the master played Re3; under μ our Ne5 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Qb2 as their Nc2, our Pg7 as their Qe2; transferred move e5c4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#3724` (ply 60) [structural distance $d=80$, QAP obj=54]: the master played Nd7+; under μ our Ne5 plays the role of that game’s Ne5; cross-type roles: our rb7 as their pa4, our pe4 as their nd6; transferred move e5c6. Synthesis: 2 retrieved analogs transferred to Nc4 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



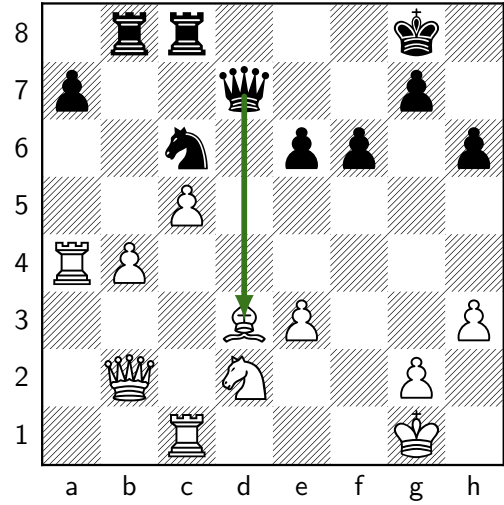
(a)



(b)

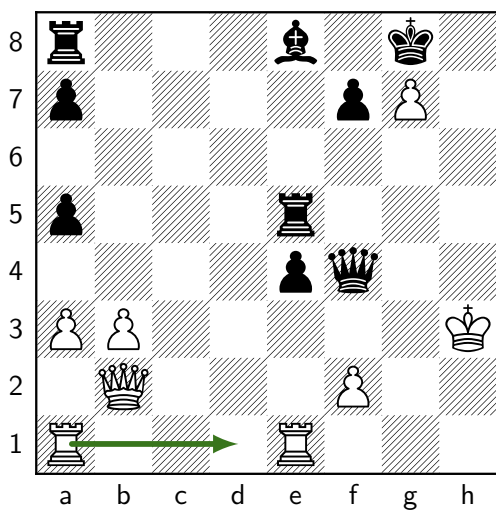


(c)

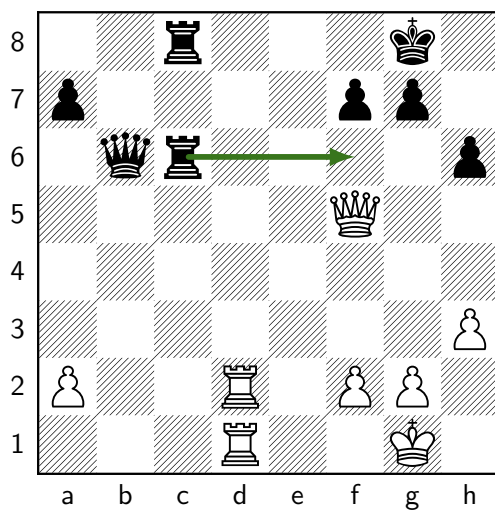


(d)

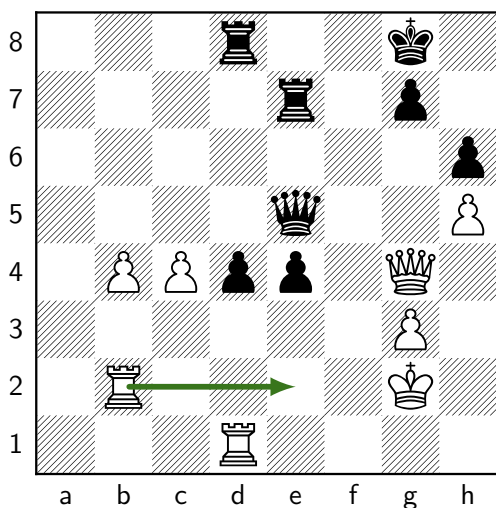
Figure 26: **Move 33, White: Ne5.** (a) Our position; Melanie plays Ne5 (c4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#3459` (ply 66) [structural distance $d=71$, QAP obj=78]: the master played Nh4; under μ our Nc4 plays the role of that game’s Nf5; cross-type roles: our pf7 as their rb2, our qg4 as their rd2; transferred move c4d2. (c) Retrieved from `lichess_elite_2024-09_p0013#2391` (ply 56) [structural distance $d=80$, QAP obj=46]: the master played Rxf7; under μ our Pf2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our pe4 as their Re1, our qg4 as their Pg2; transferred move f2f4. (d) Retrieved from `lichess_elite_2024-09_p0021#437` (ply 55) [structural distance $d=80$, QAP obj=43]: the master played Qxd3; under μ our Nc4 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Re1 as their Nc6, our Qb2 as their Ph6; transferred move c4d6. Synthesis: no retrieved analog transferred to Ne5 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



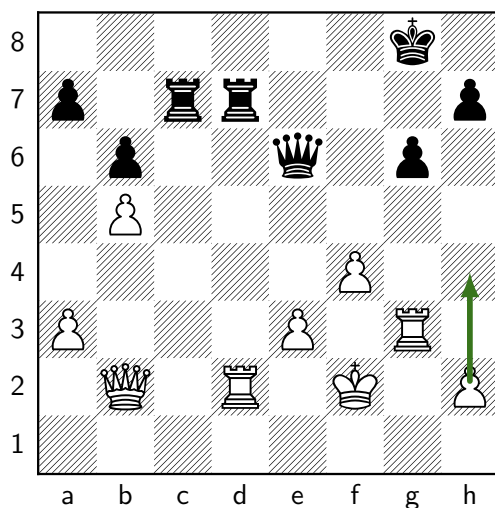
(a)



(b)

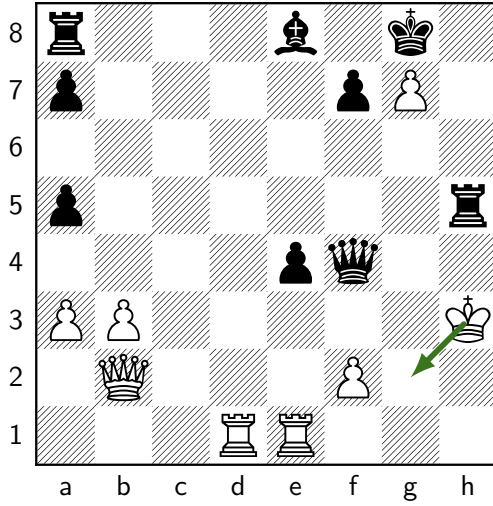


(c)

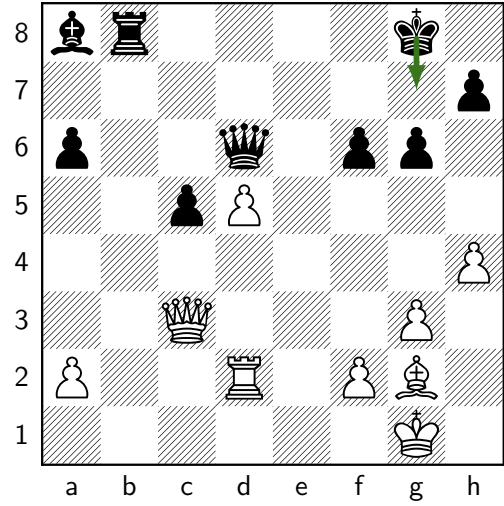


(d)

Figure 27: **Move 35, White: Rad1.** (a) Our position; Melanie plays Rad1 (a1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#463` (ply 67) [structural distance $d=87$, QAP obj=42]: the master played Rf6; under μ our *Ra1* plays the role of that game’s *Rc3*; cross-type roles: our *re5* as their *Pf7*, our *ra8* as their *Pg7*; transferred move a1d1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#2842` (ply 80) [structural distance $d=79$, QAP obj=41]: the master played Re2; under μ our *Ra1* plays the role of that game’s *Rb2*; cross-type roles: our *Qb2* as their *Pb4*, our *re5* as their *pe4*; transferred move a1d1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#3820` (ply 82) [structural distance $d=75$, QAP obj=35]: the master played h4; under μ our *Ra1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *ra8* as their *Qb2*, our *be8* as their *pa7*; transferred move a1d1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rad1 (summed QAP objective 118, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

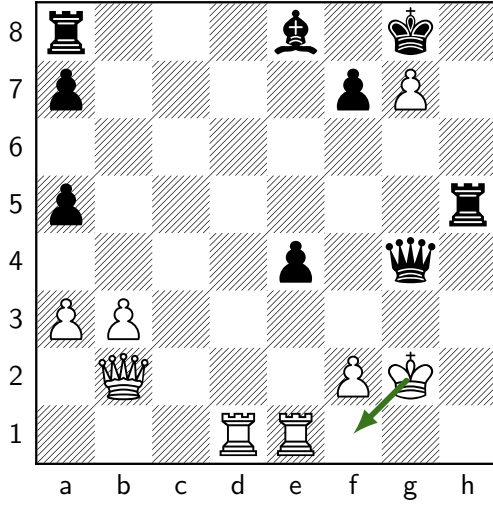


(a)

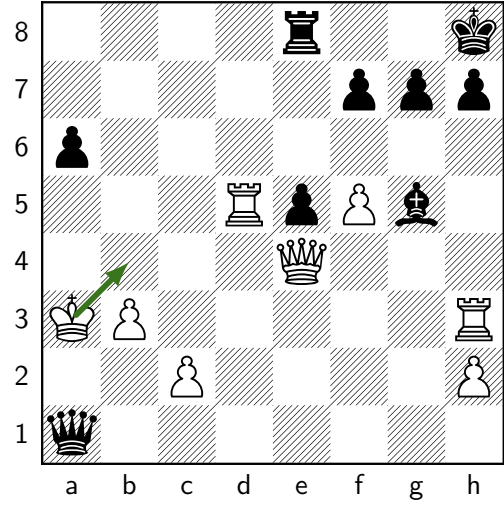


(b)

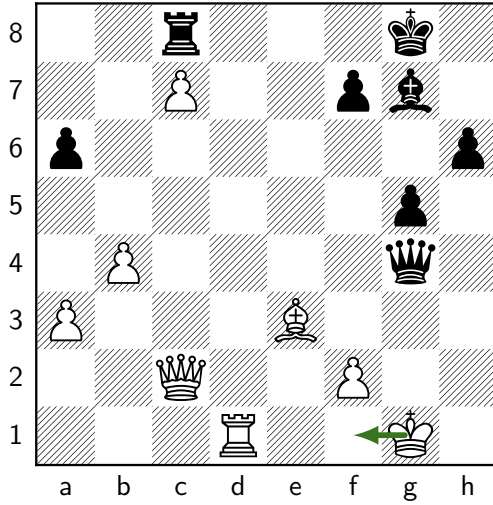
Figure 28: **Move 36, White: Kg2.** (a) Our position; Melanie plays Kg2 (**h3g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#1442` (ply 75) [structural distance $d=63$, QAP obj=17]: the master played Kg7; under μ our *Kh3* plays the role of that game’s *Kg1*; cross-type roles: our *Re1* as their *Ba8*, our *Qb2* as their *Ph7*; transferred move **h3g2 (endorses this move)**. Synthesis: 1 retrieved analog transferred to Kg2 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



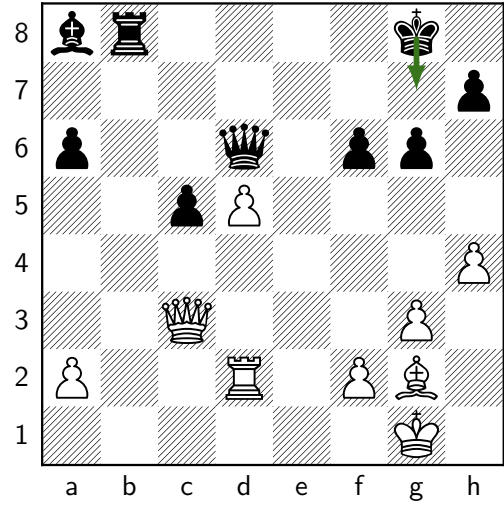
(a)



(b)



(c)



(d)

Figure 29: **Move 37, White: Kf1.** (a) Our position; Melanie plays Kf1 (g2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#1557` (ply 62) [structural distance $d=65$, QAP obj=76]: the master played Kb4; under μ our Kg2 plays the role of that game’s Ka3; cross-type roles: our rh5 as their Pb3, our Pa3 as their Rh3; transferred move g2f1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#582` (ply 62) [structural distance $d=64$, QAP obj=52]: the master played Kf1; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Pb3 as their Rd1, our rh5 as their Pb4; transferred move g2f1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0023#1442` (ply 75) [structural distance $d=62$, QAP obj=17]: the master played Kg7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Re1 as their Ba8, our be8 as their Ph7; transferred move g2f1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf1 (summed QAP objective 145, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.